

Verified Settlement Search

Settlement Semigroups, Cross-Agent Geometry,
Reflective Self-Simulation, and a Return to Conjectures

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Abstract

This manuscript develops a self-contained bridge from verified theorem memory to reflective control and then back to the practical goal of conjecture settlement. It draws together the semi-ideal computer (SIC) and automated theorem prover (ATP) framework of *Depths of Induction and Formalized Self-Awareness*, the BANK/FUTUREBANK and P/R/I/C architecture of *What Checks the Proof?*, the controlled iterated-system and fixed-point viewpoint of DATA 3.0, and the value-function/compass program of AMLD control theory. The BANK–SIC representation theorem is retained: after forgetting record metadata, an AMLD BANK trajectory is a SIC trajectory on an expanded certificate language, and the starting set Γ has the same mathematical function in both frameworks. Under explicit certificate-search hypotheses the global P/R/I/C settlement process is an ATP-like target search for a settlement proposition.

The paper continues to treat the complete verifier-gated P/R/I/C federation as one evolving discrete controlled system. The earlier Controlled Abel–Bellman Settlement Theorem and Half-Step Compass Theorem are retained explicitly as classical foundations: Bellman optimality and half-gap stability are not presented as new general mathematics. They are then strengthened by a new framework-specific theorem sequence. First, the four role updates generate outcome-labeled settlement languages whose residuals describe all future P/R/I/C settlement possibilities; this yields exact notions of cross-role catalysis, role-ablation penalty, and noncommutative role-order defect. Second, a finite-horizon branchwise distance vector produces a 4×4 cross-agent progress matrix, and quarter-step-accurate component predictions suffice to recover that integer interaction geometry exactly by rounding. Third, a checker-backed Level-2 self-model that locally semiconjugates the four concrete role maps to four internal role maps is shown, by induction on role words, to simulate every finite abstract future of the whole ATP.

The reflective theorem also yields a canonical settlement-future quotient: two global states are identified exactly when no finite P/R/I/C continuation distinguishes their verifier-labeled future behavior. Every exact checked self-model must refine this quotient, which gives a lower bound on the number of distinguishable self-states—and hence on the worst-case number of bits—required for exact Level-2 settlement self-simulation. The earlier Checked Self-State Compass Sufficiency result is retained as a derived corollary: its factorization hypothesis now follows from the stronger generator-exact self-simulation theorem rather than being assumed independently. The quotient also carries a natural four-generator growth function. For a fixed initial state, N_H counts the distinct settlement-future quotient states reached by the 4^H role words of length H . This yields a sharp escape criterion: if the exponential growth exponent of N_H has limsup strictly below 1 when normalized against 4^H , then $N_H/4^H \rightarrow 0$. A companion product bound separates the number

of realized four-coordinate outcome-future profiles from the much larger Cartesian product of coordinate possibilities. The same growth criteria admit a nonstandard formulation at unlimited hypernatural horizons, with Transfer used only after a uniform standard eventual bound has been stated.

A prior-art boundary is stated conservatively. Bellman control, action-gap robustness, generic state abstraction, Nerode-style future equivalence, agentic theorem-proving MDPs, and multi-agent proof infrastructure are cited as antecedents. The manuscript does not claim historical priority for those ingredients. The research claim isolated for further review is the exact synthesis around a verifier-gated P/R/I/C settlement computer, outcome-labeled residual future languages, cross-role settlement geometry, quarter-step exact recovery, and checker-backed self-simulation of the same machine. Three empirical conjectures—cross-role catalysis, noncommutative scheduling advantage, and reflective settlement compression—are clearly separated from proved results and are proposed as targets for sealed experiments before any application to genuinely open mathematics.

Status and novelty discipline. Results labeled theorem, proposition, or corollary are proved from the hypotheses stated here. Results labeled conjecture are not promoted to theorems. Classical ingredients are cited rather than renamed as novel. The strengthened claims concern the exact AMLD-specific constructions stated here, and even those remain appropriate subjects for independent literature and patent-database review. No open mathematical conjecture is claimed settled in this manuscript.

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1 Why another text: from checking proofs to watching search

What Checks the Proof? begins with a deliberately simple slogan:

“Imagine freely; verify rigidly.”

The point is architectural rather than rhetorical. A machine may explore aggressively, hypothesize lemmas, rank branches, or reason about its own search state, but only checker-accepted certificates enter verified mathematical memory [2]. The same text later states the stronger epistemic principle

“The machine may be uncertain about what to try next while being exact about what it has already certified.”

That distinction leaves a natural sequel question. Once verification has been separated from search, what mathematical object is doing the searching, what can it know about itself, and how should that knowledge affect the next move?

The present manuscript is written so that the reader does not need to keep earlier texts open beside it. Definitions needed from the earlier work are restated and connected rather than inserted without explanation. The intellectual line is roughly

$$\begin{aligned} \text{formal system} &\longrightarrow \text{SIC/ATP} \longrightarrow \text{BANKed AMLD} \\ &\longrightarrow \text{controlled IFS} \longrightarrow \text{reflective controller} \\ &\longrightarrow \text{verified settlement.} \end{aligned}$$

The working title *What Watches the Search?* names the reflective-control direction. Two further working titles make the longer trajectory explicit: *Initial Segments of Tegmark’s Self-Aware Substructures: Formal Self-Aware Automated Theorem Provers* would isolate the tagged/reflection theory, while *Techniques to Settle Conjectures* would deliberately shift attention back to end-to-end mathematical targets. These are working directions rather than claims that each title already denotes a finished publication. This paper sits between them: it develops enough theory to make “watching the search” useful, but it keeps returning to the question of whether the resulting machine actually reaches a certificate.

Several earlier papers supply pieces of this bridge. *Depths of Induction and Formalized Self-Awareness* formalizes SICs, ATPs, tags, self-theories, and reflection levels [3]. DATA 3.0 represents a verifier-backed conjecture-settling machine by complete state transformations forming a controlled iterated system and transformation semigroup [4]. *Proof Compass Theory* separates learnability, shrinkage, and adaptive control from their unrestricted conjectural forms [5]; the later AMLD compass-control synthesis identifies exact distance-to-settlement with a minimum-time value function, its negative with an Abel coordinate, and Bellman minimization with feedback control [6]. The *Reflective Compass Control Research Program* then proposes checked self-state as input to a safe meta-controller rather than as a new source of proof authority [7]. *Automatic Metalogical Deciders, Part Two* insists that prolonged P/R failure is not itself evidence for independence and uses fair scheduling as the canonical semantics, while treating adaptive scheduling as an experimental comparator [8]. These are not separate metaphors. Under suitable hypotheses they are views of the same machine.

2 The formal starting point and the role of Γ

2.1 Formal systems and consequence

Fix a formal presentation

$$\mathfrak{F} = (\Sigma, A, R),$$

where Σ is a formal language, A is an allowed axiom/hypothesis universe, and R is a declared family of finitary inference rules. For an admitted hypothesis set $\Gamma \subseteq A$, let

$$\text{Con}_{\mathfrak{F}}(\Gamma)$$

denote the least inference-closed set containing Γ . A finite derivation of φ from Γ is written

$$\Gamma \vdash_{\mathfrak{F}} \varphi.$$

The key fact for this paper is not a particular proof calculus. It is that Γ is fixed before search begins and supplies the base against which every derived claim is scoped.

2.2 Semi-ideal computers

The SIC framework makes the evolving theorem state explicit. The earlier definition can be paraphrased as follows [3]. A SIC over a formula domain Y is a pair $M = (C, H)$ with

$$C : \mathcal{P}(Y) \times \mathbb{N}_+ \rightarrow \mathcal{P}(Y), \quad H : \mathcal{P}(Y) \times \mathbb{N}_+ \rightarrow \{0, 1\},$$

satisfying, for every $\Gamma \subseteq Y$ and $m \in \mathbb{N}_+$,

$$C(\Gamma, 1) = \Gamma, \tag{1}$$

$$C(\Gamma, m) \subseteq C(\Gamma, m + 1), \tag{2}$$

and, once $H(\Gamma, m) = 1$, both the halting bit and the state persist:

$$H(\Gamma, m + 1) = 1, \tag{3}$$

$$C(\Gamma, m + 1) = C(\Gamma, m). \tag{4}$$

Thus a SIC is not merely an algorithm with an input string. It is a staged computation whose mathematical state begins exactly at Γ and expands monotonically until, if a designated halt occurs, the state freezes.

Depths summarizes its broad ATP notion in a sentence worth retaining:

“An automated theorem prover ... is a SIC ... intended to enumerate theorems or search for designated targets.”

The ellipses suppress the domain qualifiers, not mathematical content [3]. A target-search ATP adds a liveness condition: if the designated target is derivable, some finite stage halts with the target in the materialized state.

2.3 The BANK in an AMLD

An Automated Mathematical Logic Decider (AMLD) begins from a richer environment

$$\mathcal{E} = (\Gamma, c, \mathcal{L}, \mathcal{R}, \mathcal{C}_{\text{check}}, \nu),$$

containing the base Γ , target c , language, inference rules, declared certificate/checker families, and an environment version. Its verified BANK begins with typed records for Γ :

$$\mathcal{B}_0 = \text{seed}(\Gamma, \mathcal{E}).$$

The BANK may then grow by adding checker-accepted derived records. The extra record fields—scope, semantic certificate kind, dependencies, provenance, verifier receipt, environment identifier, and deposit time—matter for safety, but they do not change the mathematical content of an admitted axiom [2].

There is therefore a canonical forgetful map

$$\text{cont} : \{\text{BANK records}\} \rightarrow Y^\dagger,$$

where $Y^\dagger$ is an expanded language capable of representing both object-level and declared metalevel certificate statements. For a set of records B , write

$$\text{cont}(B) = \{\text{cont}(b) : b \in B\}.$$

By construction,

$$\text{cont}(\mathcal{B}_0) = \Gamma$$

when the initial metadata are forgotten.

3 The BANK–SIC bridge: settlement is computation with verified banks

The phrase “the BANK begins with Γ ” can sound like an implementation convention. In fact it is the direct analogue of the SIC axiom $C(\Gamma, 1) = \Gamma$. The following theorem makes that claim precise.

Definition 3.1 (BANK trajectory). Fix an AMLD instance $(\mathcal{E}, c)$ and write

$$\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2, \dots$$

for its verified BANK states after successive completed atomic turns. Assume:

- (B1) **Seed correctness:** $\text{cont}(\mathcal{B}_0) = \Gamma$.
- (B2) **Monotone deposit:** $\mathcal{B}_m \subseteq \mathcal{B}_{m+1}$ until terminalization.
- (B3) **Verifier gating:** every new mathematical record in $\mathcal{B}_{m+1} \setminus \mathcal{B}_m$ is admitted by the declared checker relative to its recorded scope and dependencies.
- (B4) **Terminal freezing:** once a terminal settlement certificate is accepted, later BANK states are unchanged.

Let $H_D(\Gamma, m+1) = 1$ exactly when the AMLD has terminalized by BANK stage m .

Theorem 3.2 (BANK–SIC representation). *Under (B1)–(B4), define for $m \in \mathbb{N}_+$*

$$C_D(\Gamma, m) := \text{cont}(\mathcal{B}_{m-1}).$$

Then $M_D = (C_D, H_D)$ is a SIC over the expanded declared certificate language $Y^\dagger$.

Proof. By (B1),

$$C_D(\Gamma, 1) = \text{cont}(\mathcal{B}_0) = \Gamma,$$

which is the SIC seed axiom. By (B2), $\mathcal{B}_{m-1} \subseteq \mathcal{B}_m$, hence

$$C_D(\Gamma, m) \subseteq C_D(\Gamma, m+1).$$

If $H_D(\Gamma, m) = 1$, terminalization has already occurred. By definition the halting condition remains true at every later stage, and by (B4) the BANK freezes. Therefore

$$H_D(\Gamma, m+1) = 1, \quad C_D(\Gamma, m+1) = C_D(\Gamma, m).$$

These are exactly the SIC persistence and freezing axioms. The verifier-gating condition (B3) is stronger than the bare SIC axioms: it supplies the intended semantic reason that BANK growth represents certified mathematical computation rather than arbitrary set growth. $\square$

Corollary 3.3 (Γ has the same structural function). *Modulo BANK metadata, Γ plays the same mathematical role in the SIC and AMLD frameworks: it is the admitted initial state from which the certified computational state grows, and it remains the reference base for scope and consequence throughout the run.*

Proof. In the SIC, the role is forced by $C(\Gamma, 1) = \Gamma$. In the AMLD, Theorem 3.2 identifies the metadata-forgotten seed BANK with the same stage-one state. Every later certified state extends that base. Thus the difference is representational—plain formulas versus typed records—not a difference in the logical function of Γ . $\square$

This is the sense in which settlement is *computing with banks*. A SIC materializes a monotone theorem state. An AMLD materializes a richer, provenance-aware verified state. The latter can be viewed as a typed lift of the former.

Remark 3.4 (The AMLD BANK is strictly richer data). The corollary does not identify a BANK record with a formula. An AMLD record may carry object/metalogical level, certificate body, dependencies, role provenance, scope, and checker receipt. The claim is that after applying cont , the underlying admitted mathematical state has the SIC form. This is why BANK memory can safely support cross-role reuse without confusing discovery history with logical consequence.

3.1 From SIC to ATP-like settlement search

Let the declared terminal certificate kinds be

$$P, \quad R, \quad I, \quad C,$$

standing respectively for a checked proof of c , a checked proof of $\neg c$, an accepted independence metacertificate, and an accepted inconsistency certificate for the declared base. In an expanded metatheory, introduce the settlement proposition

$$\text{Settled}_{\Gamma, c} := P_{\Gamma, c} \vee R_{\Gamma, c} \vee I_{\Gamma, c} \vee C_{\Gamma}.$$

Theorem 3.5 (Conditional settlement-ATP theorem). *Assume Theorem 3.2. Suppose in addition that:*

- (S1) *every accepted terminal certificate causes a verified BANK record entailing $\text{Settled}_{\Gamma,c}$;*
- (S2) *the declared terminal certificate families are effectively enumerable/checkable on the instance class under consideration; and*
- (S3) *the combined P/R/I/C schedule contains a persistent fair fallback that eventually examines every candidate in those declared families.*

Then M_D is a target-search ATP for $\text{Settled}_{\Gamma,c}$ relative to that declared settlement language: whenever an admissible terminal certificate exists, the combined search eventually halts with a verified settlement record.

Proof. Theorem 3.2 gives the SIC structure. By (S1), halting implies that the designated settlement proposition is represented in the materialized state. If an admissible terminal certificate exists, effective enumerability/checkability and fair persistent coverage imply that the relevant candidate is eventually examined. When examined, the checker accepts it, it is deposited, and terminalization occurs. This is exactly the target-search liveness condition for the declared settlement proposition. $\square$

Caution 3.6 (No universal decision procedure). The theorem is deliberately conditional. In arbitrary sufficiently expressive theories, independence or other metacertificate families need not be available as one recursively searchable complete class, and no scheduling trick removes undecidability. The result says that where a certificate class is effectively covered, reflective or heuristic search can be placed around it without destroying the covered guarantee.

4 The whole P/R/I/C schedule as one machine

A common notation writes the role schedule locally as

$$P(1), R(1), I(1), C(1), P(2), R(2), I(2), C(2), \dots$$

The parenthesized number can mean the local turn count of the role. If a single global clock is preferred, the same run can be written

$$P[1], R[2], I[3], C[4], P[5], R[6], \dots$$

The distinction is bookkeeping. The important point is that the schedule is not four unrelated theorem provers observed from outside. Once BANK, verifier history, resource state, FUTUREBANK, and controller state are shared, the schedule defines one evolving machine.

A convenient complete state is

$$\Sigma_m = (\mathcal{B}_m, \mathcal{FB}_m, H_m, \text{Ref}_m, \text{Ctrl}_m, q_m),$$

where q_m includes whatever phase/scheduler data are needed to determine the next role. DATA 3.0 uses a more detailed complete state containing the theory/target/metatheory triple, typed BANK, multiple search frontiers, untrusted proposal buffers, history, learned parameters, resource allocation, certificate records, status, and a round counter [4]. The mathematical requirement is Markovian completeness: the next update must be determined by the declared state together with any frozen external policy.

5 The Universal Proof-Horizon Operator and the NSA unlimited horizon

The preceding SIC discussion is about staged computation. A closely related earlier construction, the *Universal Proof-Horizon Operator*, starts from an effective presentation of proofs and asks a more concrete question: if a target is provable, how far out in an effectively ordered proof-cost search must one go before a proof appears? [1] The nonstandard-analysis chapter of *What Checks the Proof?* asks a complementary idealized question: what happens if the bound is not merely very large but larger than every standard natural? [2]

These two uses of the word *horizon* should not be conflated. The Proof-Horizon paper writes its proof-cost function as $c(p)$ and its operator as $H_{T,V,c}$. In the present manuscript, c already denotes the conjecture under settlement, so this section renames the proof-cost function κ and writes the operator as

$$\mathcal{H}_{T,V,\kappa}.$$

Likewise, the nonstandard chapter writes its unlimited bound as H . To prevent that symbol from being mistaken for the Proof-Horizon operator, this section denotes the unlimited nonstandard natural by K .

5.1 The ordinary computable horizon

An *effective proof presentation* consists, schematically, of a formal theory T , a total mechanical verifier V , and a total computable cost function

$$\kappa : \mathbb{N} \rightarrow \mathbb{N}$$

with effectively finite bounded sublevel sets: for every standard b , the set $\{p : \kappa(p) \leq b\}$ is finite and effectively listable. This is the effective-properness condition used in the Proof-Horizon paper. Let

$$\sigma_0, \sigma_1, \sigma_2, \dots$$

be a fixed computable enumeration of the sentences of the language.

The Universal Proof-Horizon theorem gives a partial computable operator $\mathcal{H}_{T,V,\kappa}(n)$ that searches the cost layers in increasing order. It halts exactly when $T \vdash \sigma_n$ and, when it halts, returns the fixed tie-broken least proof code among proofs of minimum κ -cost [1]. Define the corresponding theorem-specific finite horizon by

$$h_{T,V,\kappa}(n) := \kappa(\mathcal{H}_{T,V,\kappa}(n))$$

whenever the operator is defined. Thus $h_{T,V,\kappa}$ is partial computable on exactly the theorem indices.

This is a semidecision boundary, not a decision procedure. If σ_n is not a theorem, ordinary search need not halt. The fact that a verifier can recognize a proof when one arrives does not provide an ordinary finite test for the absence of every proof.

5.2 The nonstandard bounded proof condition

The NSA formulation needs a stronger semantic environment than an arbitrary nonstandard model of arithmetic. Expand the first-order language of the standard natural-number structure by symbols interpreting the verifier relation V and the total cost function κ (equivalently, use formulas defining their graphs), and call the resulting standard structure $\mathcal{N}_{T,V,\kappa}$. Let

$$\mathcal{N}_{T,V,\kappa} \preceq M$$

be an elementary extension. Let $K \in M$ be *unlimited*, meaning

$$K > m \quad \text{for every standard } m \in \mathbb{N}.$$

For a standard sentence φ , define the cost-bounded proof-existence condition

$$B_{K,\kappa}^M(\varphi) \quad :\Longleftrightarrow \quad M \models (\exists p)[\kappa^M(p) \leq K \wedge V^M(p, \ulcorner \varphi \urcorner) = 1]. \quad (5)$$

This is a truth condition in the elementary extension. It is not an instruction to an ordinary computer to execute K steps.

Remark 5.1 (Arithmetic extension versus full nonstandard analysis). Theorem 5.2 uses only an elementary extension of the expanded natural-number structure. A full nonstandard-analysis universe with a star map, internal sets, and hyperfinite intervals is not required. If such an enlargement is chosen, K may be viewed as an unlimited hypernatural and the bounded search may be pictured as an internal hyperfinite scan; the theorem itself depends only on elementarity and unlimitedness. This is the same framework separation enforced in the nonstandard chapter of *What Checks the Proof?* [2].

The $B_H^M(\varphi)$ notation used in the nonstandard chapter of *What Checks the Proof?* is the corresponding proof-code-bounded special case

$$M \models (\exists p)[p \leq H \wedge \text{Prf}_{\text{PA}}(p, \ulcorner \varphi \urcorner)].$$

In effect, that special case orders witnesses by the code-value bound itself. Equation (5) is the version aligned with the Proof-Horizon paper's more general proof-cost function.

Theorem 5.2 (Proof-Horizon–NSA equivalence). *Under the hypotheses above, for every standard sentence σ_n ,*

$$\boxed{\mathcal{H}_{T,V,\kappa}(n) \downarrow \quad \Longleftrightarrow \quad T \vdash \sigma_n \quad \Longleftrightarrow \quad B_{K,\kappa}^M(\sigma_n).}$$

Whenever these equivalent conditions hold,

$$h_{T,V,\kappa}(n) < K.$$

Proof. The first equivalence is the Universal Proof-Horizon theorem: the operator halts exactly on theorem indices.

Assume $T \vdash \sigma_n$. Then there is an ordinary finite proof code p_* accepted by V . Its cost $\kappa(p_*)$ is a standard natural. Since K exceeds every standard natural,

$$\kappa(p_*) < K.$$

The elementary extension agrees with the standard structure on the represented verifier and cost relations for the standard parameters, so the same standard witness p_* establishes $B_{K,\kappa}^M(\sigma_n)$.

Conversely, suppose $B_{K,\kappa}^M(\sigma_n)$ holds. Dropping the bound yields

$$M \models (\exists p) V^M(p, \ulcorner \sigma_n \urcorner) = 1.$$

Because the target code is standard and M is an elementary extension of the standard expanded structure,

$$\mathcal{N}_{T,V,\kappa} \models (\exists p) V(p, \ulcorner \sigma_n \urcorner) = 1.$$

Thus an ordinary finite T -proof of σ_n exists, so $T \vdash \sigma_n$.

Finally, when the horizon operator halts, its minimum cost $h_{T,V,\kappa}(n)$ is a standard natural, and therefore is smaller than the unlimited K . $\square$

Corollary 5.3 (Unlimited bounds dominate all standard theorem horizons). *For every standard theorem index n ,*

$$h_{T,V,\kappa}(n) < K.$$

Thus one unlimited nonstandard cost bound lies beyond every theorem-specific standard finite Proof Horizon, although it does not supply an executable universal standard cutoff.

This is the precise sense in which the NSA construction extends the horizon idea. The ordinary operator moves through

$$0, 1, 2, \dots, h_{T,V,\kappa}(n)$$

until the first minimum-cost proof layer succeeds. The NSA statement places an idealized bound K beyond every standard finite layer at once. It does not replace the theorem-specific minimum: $h_{T,V,\kappa}(n)$ remains the exact finite horizon for a provable target, while K is an external upper bound dominating all standard such horizons.

5.3 SIC stages, AMLD routes, and the same boundary

When κ is proof length and the canonical breadth-first SIC materializes exactly the formulas reachable within successive proof-length layers, its first halting stage on a theorem is the shortest proof length. Thus the SIC first-halting stage and the corresponding Proof Horizon are two descriptions of the same finite boundary in that cost model. More general effectively proper costs give analogous cost layers even when they are not identical to SIC stage count.

For conjecture settlement, the generalized bounded condition can be applied to the positive object-level channels

$$B_{K,\kappa}^M(c), \quad B_{K,\kappa}^M(\neg c), \quad B_{K,\kappa}^M(\perp).$$

They correspond, respectively, to the existence of standard finite P -, R -, and C -certificates covered by the represented verifier. If T is consistent, then the simultaneous failures

$$\neg B_{K,\kappa}^M(c) \quad \text{and} \quad \neg B_{K,\kappa}^M(\neg c)$$

are equivalent, at the metatheoretic level, to

$$T \not\vdash c \quad \text{and} \quad T \not\vdash \neg c.$$

This is a derivability-based independence classification. It must not be confused with an ordinary finite I -certificate produced by a standard AMLD. If the architecture requires finite model, relative-consistency, or other metacertificates for I , those remain separate certificate obligations. If a contradiction certificate exists, the C channel should receive the declared contradiction-dominance priority.

Caution 5.4 (The NSA condition is not a new standard decider). The equivalence

$$\neg B_{K,\kappa}^M(\varphi) \iff T \not\vdash \varphi$$

is a semantic consequence of the chosen elementary extension on standard targets. It does not give a standard computer an effective test for the divergence of $\mathcal{H}_{T,V,\kappa}(n)$. Compactness, elementarity, and unlimitedness provide a model-theoretic idealization, not an ordinary finite runtime. This preserves the Church–Gödel boundary of the Proof-Horizon construction.

The resulting lineage is therefore

$$\boxed{\text{Proof Horizon} \longrightarrow \text{SIC horizon} \longrightarrow \text{NSA unlimited horizon} \longrightarrow \text{P/R/I/C settlement semantics.}}$$

The first two are ordinary staged/computable search notions under their stated hypotheses. The third is a nonstandard semantic idealization. DATA 3.0 then supplies the dynamical question that remains after the horizon is characterized: how should the evolving complete state be controlled so that an actual verifier-gated run reaches the relevant terminal fixed-point class efficiently?

6 DATA 3.0: the search is a controlled discrete iterated system

DATA 3.0 states the central point explicitly:

“Data 3.0 is more accurately a controlled or place-dependent iterated system than a classical contractive Hutchinson IFS.”

This sentence is important because it separates two claims that are often conflated [4]. The first is structural: the machine evolves by repeatedly selecting state maps. The second is metric: those maps might or might not be contractions. The structural claim is immediate once complete state is fixed; the contraction claim requires a separate theorem and, on the unquotiented state space, is generally false.

Let X be the complete state space. For each active subsystem or role j , let

$$F_j : X \rightarrow X$$

be the realized verified state transformation. The map can include proposal, verification, BANK/history update, and controller bookkeeping. A deterministic controller π gives

$$j_n = \pi(x_n), \quad x_{n+1} = F_{j_n}(x_n).$$

Finite compositions generate a transformation semigroup

$$S = \langle F_j : j \in J \rangle \subseteq X^X.$$

The semigroup need not commute: changing the order of two search/control transformations generally changes which BANK and history the second one sees.

Theorem 6.1 (Controlled discrete IFS representation). *Any deterministic AMLD implementation whose complete state contains every variable needed for the next update and whose admissible atomic subsystems act by maps $F_j : X \rightarrow X$ is a discrete-time controlled/place-dependent iterated system. Under a fixed deterministic controller π , its orbit is the iteration of the state-dependent map family according to $j_n = \pi(x_n)$.*

Proof. At each discrete time n , the controller selects one index j_n from the declared family and the next complete state is $F_{j_n}(x_n)$. This is exactly iteration of a family of state transformations with place-dependent control. No contractivity or fractal-attractor hypothesis is needed for this representation. $\square$

6.1 Round robin can be made autonomous

For a strict four-role cyclic schedule, augment the state by

$$q \in \mathbb{Z}/4\mathbb{Z}$$

and define $a(0) = P$, $a(1) = R$, $a(2) = I$, $a(3) = C$. On

$$\tilde{X} = X \times \mathbb{Z}_4$$

define

$$T(x, q) = (F_{a(q)}(x), q + 1 \pmod{4}).$$

Then iterating a single map T reproduces the cyclic P/R/I/C schedule. One full round may alternatively be packaged as a macro-update

$$T_4 = \text{Terminalize} \circ F_C \circ F_I \circ F_R \circ F_P,$$

with terminal checks inserted after atomic actions when immediate freezing is required.

This conversion is useful later: controlled search can be studied either as a semigroup of available maps or, after freezing a policy/schedule, as an autonomous discrete dynamical system.

7 Settlement as fixed-point geometry

Let $\mathcal{C} \subseteq X$ be the set of states containing an accepted terminal certificate and terminal status. The DATA 3.0 construction uses terminalization so that accepted settlement states are absorbing. Under a macro-update T , its fixed-point theorem takes the form

$$\text{Fix}(T) = \mathcal{C}$$

when every nonterminal macro-round changes a monotone counter and every terminal state thereafter remains unchanged [4]. The counter rules out nonterminal cycles in the complete state.

The fixed-point identity is a stopping-state theorem, not a reachability theorem. It says what settlement looks like if it happens. It does not say that every orbit reaches settlement. This distinction will matter throughout the paper.

8 A refutation: the complete AMLD is not a Hutchinson contractive IFS

Hutchinson's classical IFS framework uses a finite family of contractions on a complete metric space [12]. DATA 3.0 deliberately did not claim this stronger property. The terminal geometry now yields a direct obstruction.

Theorem 8.1 (Full-state noncontractivity). *Let (X, d) be any metric realization of the complete AMLD state space. Suppose two distinct certified terminal states $x_*, y_* \in \mathcal{C}$ are left unchanged by every subsystem map F_j . Then no F_j is a strict contraction on X . Consequently the family $\{F_j\}$ is not a Hutchinson-style contractive IFS on the complete unquotiented state space.*

Proof. Fix j . Since terminal states freeze,

$$F_j(x_*) = x_* \quad \text{and} \quad F_j(y_*) = y_*.$$

If F_j were a strict contraction with constant $0 \leq q < 1$, then

$$d(x_*, y_*) = d(F_j x_*, F_j y_*) \leq q d(x_*, y_*).$$

Because $x_* \neq y_*$, the metric gives $d(x_*, y_*) > 0$. Division yields $1 \leq q$, contradiction. $\square$

Corollary 8.2 (Macro-map obstruction). *If a deterministic settlement macro-map T has at least two distinct terminal fixed states, then T cannot be a strict contraction under any genuine metric on the complete unquotiented state space.*

The theorem is stronger than a failed search for the right metric. As long as different terminal states remain metrically distinct and are all fixed, no metric can make a shared terminal-freezing map strictly contractive.

Globally, a multi-target architecture naturally has many distinct terminal states because the target is part of complete state. Within a single target fiber there may also be multiple frozen terminal states corresponding to different accepted certificates or histories, unless terminalization deliberately canonicalizes them into one state.

9 A complementary theorem: terminal quotienting can create a contraction

The noncontractivity theorem suggests a natural repair: identify all terminal states. Surprisingly, on a deterministic basin in which every state settles after finitely many steps, this repair is enough to construct an explicit contraction metric.

Definition 9.1 (Finite-settling basin). For a deterministic map $T : X \rightarrow X$ with terminal fixed set $\mathcal{C}$, define

$$B := \{x \in X : \exists n \geq 0 \text{ with } T^n x \in \mathcal{C}\}.$$

For $x \in B$, define the remaining hitting time

$$r(x) := \min\{n \geq 0 : T^n x \in \mathcal{C}\}.$$

Thus $r(x) = 0$ exactly on $\mathcal{C}$.

Identify all terminal states and no others:

$$x \sim y \iff x = y \text{ or } (x, y \in \mathcal{C}).$$

Let

$$\bar{B} = B / \sim$$

and denote the common terminal class by $*$. Since T fixes $\mathcal{C}$, it induces a well-defined map

$$\bar{T} : \bar{B} \rightarrow \bar{B}.$$

For a nonterminal class $\bar{x}$ define $r(\bar{x}) = r(x)$ and set $r(*) = 0$.

Lemma 9.2 (Depth drops by one). *If $\bar{x} \neq *$, then*

$$r(\bar{T}\bar{x}) = r(\bar{x}) - 1.$$

Proof. If $r(\bar{x}) = m > 0$, then $T^m x$ is terminal and no earlier iterate is. Starting from Tx , the first terminal iterate therefore occurs after exactly $m - 1$ further steps. $\square$

Theorem 9.3 (Quotient contractivization). *Fix any $q \in (0, 1)$. On $\bar{B}$, join every nonterminal vertex v to its parent $\bar{T}v$. Give that edge length*

$$w(v, \bar{T}v) := q^{-r(v)}.$$

Let d_q be the induced path metric. Then $(\bar{B}, d_q)$ is a complete metric space and

$$d_q(\bar{T}x, \bar{T}y) \leq q d_q(x, y)$$

for all $x, y \in \bar{B}$. Hence $\bar{T}$ is a strict contraction with unique fixed point $$.*

Proof. Every vertex has a unique parent and a finite directed path to $*$. The underlying undirected graph is connected. It is acyclic: if an undirected cycle existed, choose a vertex of maximal depth r on the cycle. Both cycle-neighbors would have to lie one level closer to $*$, but a vertex has only one parent, contradiction. Thus the graph is a tree and the path metric is well-defined.

Consider an edge from a vertex v of depth $k = r(v)$ to its parent. If $k = 1$, both edge endpoints map to $*$, so the image length is 0. If $k \geq 2$, the image is the parent edge at depth $k - 1$, whose weight is

$$q^{-(k-1)} = q q^{-k}.$$

Therefore applying $\bar{T}$ to any path produces a walk whose total length is at most q times the original path length. Taking the shortest path between x and y gives

$$d_q(\bar{T}x, \bar{T}y) \leq q d_q(x, y).$$

Finally, every nonzero edge has length at least $q^{-1} > 0$. Hence every Cauchy sequence is eventually constant, so the metric is complete. A strict contraction on a complete metric space has a unique fixed point; here that point is the collapsed terminal class $*$. $\square$

Remark 9.4 (What this does and does not contract). Theorem 9.3 applies to a frozen deterministic settlement map on its finite-settling basin after terminal quotienting. It does not imply that every individual control map F_j in the original multi-map system is contractive under one common metric. Thus it does not undo Theorem 8.1; it shows that the obstruction is partly representational and disappears after quotienting the very multiplicity that caused it.

This pair of theorems gives a useful two-level picture:

complete verifier-preserving state: noncontractive

while

terminal-quotiented deterministic settling basin: contractivizable.

10 Abel and Schröder coordinates arise from settlement time

Lemma 9.2 already has the form of a functional equation. The conventional Abel equation for iteration is

$$h(Tx) = h(x) + 1.$$

Define

$$h(x) := -r(x)$$

on nonterminal points of the finite-settling basin. Then

$$h(Tx) = -r(Tx) = -(r(x) - 1) = h(x) + 1.$$

Thus the remaining settlement time is not merely analogous to an Abel coordinate; its negative is an exact Abel coordinate before the absorbing boundary.

Proposition 10.1 (Hitting-time Abel coordinate). *On any deterministic finite-settling basin,*

$$h = -r$$

satisfies

$$h(Tx) = h(x) + 1$$

for every nonterminal state x .

The terminal state itself cannot carry a finite Abel translation coordinate: if $Tx_* = x_*$, then Abel's equation would demand $h(x_*) = h(x_*) + 1$. The coordinate therefore belongs naturally to the pre-settlement basin and terminates at an absorbing boundary, exactly as emphasized in DATA 3.0 [4].

Exponentiation turns translation into multiplication. For $q \in (0, 1)$ define

$$\psi(x) = q^{-r(x)} = q^{h(x)}.$$

Then before settlement,

$$\psi(Tx) = q^{-(r(x)-1)} = q q^{-r(x)} = q\psi(x).$$

This is a Schröder-type equation. The same factor q that appears in the quotient contraction metric therefore appears as the multiplicative eigenvalue of the exponentiated Abel coordinate. The edge weight in Theorem 9.3 is precisely $\psi(v)$ at the deeper endpoint.

The chain is consequently exact:

remaining settlement time $\rightarrow$ Abel translation $\rightarrow$ Schröder scaling $\rightarrow$ contraction metric
--

11 Classical foundation I: optimal remaining settlement and Abel–Bellman control

The retrospective coordinate $r(x)$ depends on an already-fixed policy. Conjecture settlement needs a prospective question: among the moves available now, which move minimizes the remaining cost to an accepted certificate? This section keeps the original Abel–Bellman result as foundational machinery rather than treating Bellman optimality itself as the novel part of the theory.

Let $A(x)$ be the finite or countable set of admissible control actions at a nonterminal state x . An action may mean giving the next quantum to P, R, I, or C; selecting a theorem-search tunnel; proving a reusable lemma; invoking a representation change; or choosing among other verifier-neutral search transformations. For simplicity first give every action unit cost and write

$$F_a(x)$$

for the successor state.

Definition 11.1 (Optimal remaining settlement time). On states from which verified settlement is reachable, define

$$r^*(x) = \min_{\pi} \{\text{number of further controlled steps needed by policy } \pi \text{ to reach } \mathcal{C}\}.$$

Set $r^*(x) = 0$ on $\mathcal{C}$ and define

$$h^*(x) := -r^*(x).$$

Theorem 11.2 (Controlled Abel–Bellman settlement theorem). *For every nonterminal state with finite r^* ,*

$$r^*(x) = 1 + \min_{a \in A(x)} r^*(F_a x),$$

and equivalently

$$\boxed{\max_{a \in A(x)} h^*(F_a x) = h^*(x) + 1.}$$

Any action

$$a^*(x) \in \arg \max_{a \in A(x)} h^*(F_a x)$$

is a shortest-settlement action.

Proof. Any successful policy from x must take one admissible first action and then complete settlement from the successor. Therefore optimality has the Bellman decomposition

$$r^*(x) = 1 + \min_a r^*(F_a x).$$

Negating and using $-\min u_a = \max(-u_a)$ yields

$$h^*(x) = -1 + \max_a h^*(F_a x),$$

which is the displayed controlled Abel equation. An action attaining the maximum attains the minimum successor hitting time and is therefore shortest-settlement. $\square$

This theorem unifies three vocabularies. In dynamic programming, r^* is a unit-cost minimum-time value function [13]. In proof-ocean language, it is exact distance-to-settlement. In iteration theory, $-r^*$ is the coordinate in which an optimal action advances translation by exactly one. Recent theorem-proving work also uses finite-horizon MDP/Bellman structure, so the Bellman recurrence itself is not presented here as a priority claim [17]. The AMLD-specific question is what additional structure becomes visible when the four settlement roles are treated as generators of one evolving verifier-gated computer.

12 Classical foundation II: the half-step compass guarantee

An exact h^* is usually as difficult to compute as the remaining search itself. The practical question is whether an approximation can still choose the correct next move. Discreteness supplies a sharp sufficient condition. The result is retained because it becomes a corollary engine for the stronger cross-agent geometry below; the underlying half-gap mechanism is classical in approximate control and action-gap analysis [14].

Theorem 12.1 (Half-step compass theorem). *Assume the unit-cost model of Theorem 11.2. Let $\hat{h}$ be an approximate compass. Suppose that at every state encountered, for every candidate successor $y = F_ax$,*

$$|\hat{h}(y) - h^*(y)| < \frac{1}{2}.$$

Then any greedy controller

$$\hat{\pi}(x) \in \arg \max_{a \in A(x)} \hat{h}(F_ax)$$

selects a shortest-settlement action. Starting from x_0 , it reaches settlement in exactly $r^(x_0)$ steps.*

Proof. Let x be nonterminal. An optimal successor y^* satisfies

$$r^*(y^*) = r^*(x) - 1,$$

so

$$h^*(y^*) = h^*(x) + 1.$$

Any nonoptimal successor z cannot have the same remaining depth, because then it would also be optimal. Since r^* is integer-valued,

$$r^*(z) \geq r^*(x),$$

hence

$$h^*(z) \leq h^*(x).$$

Thus the true compass gap between any optimal successor and any nonoptimal successor is at least 1. The strict error bound gives

$$\hat{h}(y^*) > h^*(x) + \frac{1}{2}$$

while

$$\hat{h}(z) < h^*(x) + \frac{1}{2}.$$

Therefore no nonoptimal successor can tie or outrank all optimal successors under $\hat{h}$. The greedy rule chooses an optimal action. Repeating the argument at each subsequent state gives a trajectory of optimal remaining depth, so settlement occurs after exactly $r^*(x_0)$ steps. $\square$

The threshold $1/2$ comes from the unit spacing of integer settlement depths. With unequal action costs, the analogous statement is an action-gap condition: the perturbation must be less than half the true separation between the competing action values.

Corollary 12.2 (Local ranking, not global perfection). *In the unit-cost model, a learned compass does not need globally exact values to reproduce shortest-settlement control. It is sufficient to preserve the ordering between optimal and nonoptimal immediate successors; the uniform $< 1/2$ condition is one simple sufficient guarantee.*

13 Prior-art boundary for the strengthened theorem sequence

The purpose of the next three results is not to rename classical material. Bellman optimality is classical [13]; action-gap robustness has an established reinforcement-learning literature [14]; state abstraction and sufficient representations for MDP control are established topics [15]; and future-equivalence/minimal-state constructions have a clear Nerode-style ancestry [16]. Recent theorem-proving work also models agentic proving by finite-horizon MDPs [17], while recent multi-agent theorem-proving work studies allocation and shared public proof infrastructure [18].

The research object isolated here is more specific. The whole verifier-gated P/R/I/C federation is treated as one discrete controlled machine; the four role updates generate outcome-labeled future languages; off-diagonal role effects are measured in verifier-defined settlement distance; approximate predictions are used to reconstruct an exact local interaction geometry; and a checker-backed self-model is required to simulate the generators of that same machine. These exact combinations should still be subjected to independent literature and patent-database review before any historical priority claim is made.

14 Main theorem I: settlement semigroups, residual futures, and cross-role catalysis

The four roles are now treated as generators of one computer rather than merely as four parallel programs. Freeze the role-internal policies for one control quantum, and include any random seed, resource counters, BANK contents, verifier history, and controller bookkeeping needed to make the global state complete. Let

$$J = \{P, R, I, C\}.$$

For each $j \in J$, let

$$F_j : X \rightarrow X$$

be the totalized global update produced by giving the next quantum to role j . Terminal states are frozen, so every F_j remains defined after settlement.

Let the verifier-defined terminal set be partitioned into labeled outcome classes

$$\mathcal{C} = \mathcal{C}_P \sqcup \mathcal{C}_R \sqcup \mathcal{C}_I \sqcup \mathcal{C}_C,$$

where the labels mean proof, refutation, accepted independence certificate, and accepted inconsistency certificate respectively. For a word $w \in J^*$ define $F_\epsilon = \text{id}$ and recursively

$$F_{wj} = F_j \circ F_w.$$

Thus the letters of w are applied from left to right.

Definition 14.1 (Outcome-labeled settlement language). For $k \in J$ and $x \in X$, define

$$\mathcal{L}_k(x) := \{w \in J^* : F_w(x) \in \mathcal{C}_k\}.$$

When this set is nonempty define the branchwise settlement depth

$$r_k(x) := \min\{|w| : w \in \mathcal{L}_k(x)\};$$

otherwise set $r_k(x) = \infty$.

For a word u , write

$$u^{-1}\mathcal{L} := \{v \in J^* : uv \in \mathcal{L}\}$$

for the usual left residual of a language.

Definition 14.2 (Cross-role catalysis, ablation penalty, and order defect). Assume $0 < r_k(x) < \infty$.

(a) A role $j \neq k$ is *k-catalytic* at x if

$$r_k(F_j x) = r_k(x) - 1.$$

(b) Let $r_k^{(-j)}(x)$ be the shortest length of a k -settling word over the reduced alphabet $J \setminus \{j\}$, with value ∞ if none exists. Define

$$\Delta_{j \rightarrow k}(x) := \begin{cases} r_k^{(-j)}(x) - r_k(x), & r_k^{(-j)}(x) < \infty, \\ +\infty, & r_k^{(-j)}(x) = \infty. \end{cases}$$

(c) Whenever both terms are finite, define the two-role order defect

$$\chi_{ij \rightarrow k}(x) := r_k(F_j F_i x) - r_k(F_i F_j x).$$

Theorem 14.3 (Outcome-labeled settlement-semigroup and catalysis theorem). *Under the complete deterministic unit-cost setup above, for every $x \in X$, $u \in J^*$, and $k \in J$:*

(i) **Residual-future law:**

$$\mathcal{L}_k(F_u x) = u^{-1}\mathcal{L}_k(x).$$

Thus applying a role word updates the remaining settlement language by taking a residual.

(ii) **Branchwise Bellman law:** if $x \notin \mathcal{C}$ and $r_k(x) < \infty$, then

$$r_k(x) = 1 + \min_{j \in J} r_k(F_j x).$$

Equivalently, on the finite k -basin, $h_k := -r_k$ satisfies

$$\max_{j \in J} h_k(F_j x) = h_k(x) + 1.$$

(iii) **Catalysis criterion:** a role $j \neq k$ is *k-catalytic* at x if and only if at least one shortest k -settlement word begins with j .

(iv) **Ablation criterion:**

$$\Delta_{j \rightarrow k}(x) > 0$$

if and only if every shortest k -settlement word uses role j at least once. In particular, $\Delta_{j \rightarrow k}(x) = +\infty$ means that role j is indispensable for reaching $\mathcal{C}_k$ from x within this frozen machine.

(v) **Noncommutative settlement effect:** if $F_i F_j x = F_j F_i x$, then $\chi_{ij \rightarrow k}(x) = 0$ whenever the defect is defined. Hence $\chi_{ij \rightarrow k}(x) \neq 0$ certifies that the two role orders are distinguishable by remaining k -settlement distance, even if both orders ultimately permit the same terminal outcome.

Proof. For (i), a word v belongs to $\mathcal{L}_k(F_u x)$ exactly when

$$F_v(F_u x) \in \mathcal{C}_k.$$

By the recursive composition convention this state is $F_{uv}x$, so the condition is equivalent to $uv \in \mathcal{L}_k(x)$, which is exactly $v \in u^{-1}\mathcal{L}_k(x)$.

For (ii), let $m = r_k(x)$. Every shortest settling word has a first letter j followed by a suffix of length $m - 1$ that settles F_jx into $\mathcal{C}_k$, so

$$\min_j r_k(F_jx) \leq m - 1.$$

Conversely, if some successor had $r_k(F_jx) \leq m - 2$, prefixing the corresponding word by j would settle x in at most $m - 1$ steps, contradicting minimality. Therefore the minimum successor depth is exactly $m - 1$. Negation gives the branchwise Abel form.

For (iii), the equality $r_k(F_jx) = r_k(x) - 1$ is equivalent to the existence of a length- $r_k(x) - 1$ settling suffix from F_jx ; prefixing by j gives a shortest word from x . The converse follows by removing the first letter from a shortest word.

For (iv), $\Delta_{j \rightarrow k}(x) > 0$ means that no word avoiding j has length $r_k(x)$ and reaches $\mathcal{C}_k$. This is equivalent to saying that every shortest k -settlement word contains j . If no j -free settling word exists at all, the stated value is $+\infty$.

For (v), equality of the two concrete two-step states forces equality of every state function evaluated there, including r_k . Taking the difference gives zero. Therefore a nonzero defect is a certificate of order-sensitive settlement geometry. $\square$

Corollary 14.4 (The original Abel–Bellman value is the shortest-word shadow). *In the four-role macro-action model where the top-level admissible actions are exactly J (with lower-level theorem-search, lemma, and representation choices internalized into the role maps), define*

$$r^*(x) = \min_{k \in J} r_k(x).$$

Then Theorem 11.2 is recovered by forgetting the outcome label and retaining only the length of the shortest word in

$$\mathcal{L}_P(x) \cup \mathcal{L}_R(x) \cup \mathcal{L}_I(x) \cup \mathcal{L}_C(x).$$

Thus the scalar Bellman value is a projection of the richer outcome-labeled settlement-semigroup structure.

The off-diagonal quantities are the operational point. If, for example, $\Delta_{R \rightarrow P}(x) > 0$, then every shortest proof route uses at least one quantum of the refuter. This can occur because the refuter discovers a verified lemma, exposes a failed dependency, or changes a shared representation that the prover later exploits. The architecture is then useful not merely because four things run in parallel, but because the four generators alter one another’s future settlement geometry through the shared verified state.

15 Main theorem II: quarter-step recovery of cross-agent settlement geometry

The scalar half-step theorem asks only whether the next action is optimal for the nearest settlement. A four-role Level-2 ATP can ask a richer question: what does each possible next role do to the distance of *each* terminal class?

Fix a finite planning horizon $H \in \mathbb{N}$. For $k \in J$ define the truncated branchwise depth

$$d_{k,H}(x) := \min\left(\{|w| : |w| \leq H, F_wx \in \mathcal{C}_k\} \cup \{H + 1\}\right).$$

Thus $d_{k,H}(x) \in \{0, 1, \dots, H, H+1\}$, and $H+1$ means only that no k -certificate is reachable within the declared horizon.

For $H \geq 1$ define the *cross-agent progress matrix*

$$G^H(x) = (G_{jk}^H(x))_{j,k \in J}, \quad G_{jk}^H(x) := d_{k,H}(x) - d_{k,H-1}(F_j x).$$

Rows correspond to the role receiving the next quantum; columns correspond to the eventual settlement class. Off-diagonal entries therefore measure one role's immediate effect on another role's nominal terminal objective.

For $H \geq 2$ define the horizon- H order defect

$$\chi_{ij \rightarrow k}^H(x) := d_{k,H-2}(F_j F_i x) - d_{k,H-2}(F_i F_j x).$$

Theorem 15.1 (Quarter-step cross-agent geometry recovery theorem). *Let $H \geq 1$. Suppose real-valued estimates $\hat{d}_{k,H}(x)$ and $\hat{d}_{k,H-1}(F_j x)$ satisfy*

$$|\hat{d}_{k,H}(x) - d_{k,H}(x)| < \varepsilon_0, \quad |\hat{d}_{k,H-1}(F_j x) - d_{k,H-1}(F_j x)| < \varepsilon_1$$

for every $j, k \in J$, with

$$\varepsilon_0 + \varepsilon_1 < \frac{1}{2}.$$

Define

$$\hat{G}_{jk}^H(x) := \hat{d}_{k,H}(x) - \hat{d}_{k,H-1}(F_j x).$$

Then

$$\boxed{G_{jk}^H(x) = \text{round}(\hat{G}_{jk}^H(x))}$$

for every j, k . In particular, the symmetric sufficient condition

$$\boxed{|\hat{d} - d| < \frac{1}{4}}$$

at the current state and all four one-step successors exactly reconstructs the entire 4×4 integer progress matrix.

If $d_{k,H}(x) \leq H$, then

$$G_{jk}^H(x) = 1$$

if and only if giving the next quantum to j begins a shortest horizon- H route from x to $\mathcal{C}_k$.

If $H \geq 2$ and estimates of the two residual depths in $\chi_{ij \rightarrow k}^H(x)$ are each accurate to $< 1/4$, then

$$\boxed{\chi_{ij \rightarrow k}^H(x) = \text{round}(\hat{d}_{k,H-2}(F_j F_i x) - \hat{d}_{k,H-2}(F_i F_j x))}.$$

Thus quarter-step-accurate branchwise predictions recover not only the correct next role but the exact local cross-role and order-sensitive integer geometry.

Proof. Every $d_{k,H}$ is integer-valued. By the triangle inequality,

$$|\hat{G}_{jk}^H(x) - G_{jk}^H(x)| \leq |\hat{d}_{k,H}(x) - d_{k,H}(x)| + |\hat{d}_{k,H-1}(F_j x) - d_{k,H-1}(F_j x)| < \frac{1}{2}.$$

An integer is the unique nearest integer to any real number lying strictly within $1/2$ of it, proving the rounding identity. The symmetric $1/4$ statement is the special case $\varepsilon_0 = \varepsilon_1 < 1/4$.

Now suppose $m = d_{k,H}(x) \leq H$. Any successor $F_j x$ satisfies

$$d_{k,H-1}(F_j x) \geq m - 1,$$

for otherwise prefixing a shorter successor route by j would contradict the definition of m . Equality holds exactly when j begins a shortest route of total length m . Therefore

$$G_{jk}^H(x) = m - d_{k,H-1}(F_j x) = 1$$

exactly for the shortest-route first moves.

For the two-role order defect, subtract the two approximate residual depths. The total error is again strictly less than $1/2$, while the true difference is an integer. Rounding therefore recovers $\chi_{ij \rightarrow k}^H(x)$ exactly. $\square$

Corollary 15.2 (Exact catalytic decoding from approximate values). *Under the hypotheses of Theorem 15.1, every off-diagonal entry*

$$G_{jk}^H(x) = 1, \quad j \neq k,$$

is exactly identifiable from the approximate compass and certifies that role j is a shortest-route first move toward settlement class k within horizon H .

Remark 15.3 (Weighted extension). The number $1/4$ is not metaphysical. It appears because an integer-valued progress quantity is formed by subtracting two separately estimated integer-valued depths. More generally, if the true progress values lie on a lattice with spacing $\delta > 0$, the same argument uses total two-point error $< \delta/2$; with two symmetric error budgets, $< \delta/4$ each suffices. For arbitrary nonlattice costs the appropriate replacement is a local margin condition.

16 BANKs are part of the control state, not just proof storage

A verified lemma can be logically conservative and computationally decisive. If L is already derivable from Γ , adding a checked record of L to the BANK does not strengthen the consequence relation:

$$\text{Con}(\Gamma \cup \{L\}) = \text{Con}(\Gamma).$$

But it can change the operational state because a search process no longer needs to rediscover L .

This is why the value state should be written schematically as

$$V^*(s, \mathcal{B}, \rho, \dots),$$

not merely $V^*(s)$. The earlier AMLD control synthesis makes this point directly: a theorem shell can alter the future geometry of search even when it does not strengthen the underlying theory [6]. *Automatic Metalogical Deciders, Part Two* similarly distinguishes final-proof membership from downstream BANK value: a lemma absent from one final certificate may still save substantial work on later P/R/I routes [8].

Definition 16.1 (Marginal BANK value). For a verified candidate lemma L , define its state-relative operational value under a chosen cost model by

$$\Delta_L(x) := r^*(x) - r^*(x \oplus L),$$

where $x \oplus L$ denotes the same complete state with a verifier-accepted reusable record for L added to the admissible BANK.

This quantity need not be positive in every implementation: storing or indexing a useless lemma can incur overhead. But the definition states the correct question. A lemma is valuable when its verified availability collapses future settlement cost, not merely because it is elegant or because it appeared in a previous final proof.

17 Formal self-description: what the machine can prove about itself

17.1 From Tegmark’s SAS idea to tagged ATPs

Max Tegmark’s discussion of self-aware substructures (SASs) is cosmological, not an ATP specification. The earlier self-awareness paper quotes his implicit criterion:

“a substructure is self-aware if it thinks that it is self-aware”

and interprets only the formal shape of self-ascription, not subjective consciousness [11, 3]. The theorem-proving specialization replaces phenomenology with a tag and formal statements about the tagged machine.

Let D be the global settlement machine and let

$$\langle D \rangle$$

be its designated code/name in a declared metatheory T . A formula $\varphi(\langle D \rangle)$ is self-descriptive when its intended subject is that tag. Define

$$\text{Self}(D) = \{\varphi(\langle D \rangle) : D \vdash_T \varphi(\langle D \rangle)\}.$$

A minimal Level-1 statement in the *Depths* sense is

$$D \vdash_T \text{ATP}(\langle D \rangle).$$

The point is not that a symbol can refer to itself by magic. The specification, coding, metatheory, and proof checker must make the assertion a genuine formal theorem.

17.2 Can the settlement machine prove that it is an ATP?

The external theorem has already been proved conditionally: Theorem 3.5 gives precise hypotheses under which the global BANKed settlement machine is a target-search ATP. Turning that external proof into the machine’s own self-theory requires an internalization step.

Theorem 17.1 (Level-1 self-description capability). *Let D be a global settlement machine satisfying Theorem 3.5. Let T be an effectively axiomatized metatheory capable of representing the frozen specification of D , its tag $\langle D \rangle$, and the predicate $\text{ATP}(x)$. Assume:*

- (R1) *there is a finite T -proof, from checker-admitted specification facts S_D , of $\text{ATP}(\langle D \rangle)$;*
- (R2) *the relevant proof-search component of D fairly enumerates T -proof candidates while preserving progress; and*
- (R3) *the intrinsic verifier accepts exactly the valid declared T -proof certificates for this self-description channel.*

Then, after finitely many effective self-description search steps,

$$D \vdash_T \text{ATP}(\langle D \rangle).$$

Hence D has Level-1 formal self-awareness in the tagged-ATP sense of Depths.

Proof. By (R1), a finite valid certificate exists. By fair effective enumeration with persistent progress, (R2) guarantees that the certificate is eventually proposed. By (R3), the verifier accepts it and the corresponding self-descriptive theorem enters the verified state. Therefore D proves the displayed formula about its own tag. $\square$

Caution 17.2 (Capability is not yet a concrete implementation certificate). The theorem gives a constructive route to Level 1, but an actual software release should not be called Level 1 merely because the mathematical architecture could support such a proof. The concrete tagged specification and checker-accepted proof of $\text{ATP}(\langle D \rangle)$ should be produced and archived.

17.3 Reflection depth and checked state-awareness are different

Depths defines a finite reflection sequence

$$\theta_0 = \text{ATP}(\langle D \rangle), \quad \theta_{k+1} = \text{Pr}(\langle D \rangle, \langle \theta_k \rangle),$$

and grades reflection by which terms the machine proves [3]. *What Checks the Proof?*, however, deliberately uses a different architectural scale for checked self-description. Its Level 2 includes verified dynamic facts such as the current BANK identifier, FUTUREBANK dependencies, verifier-history summaries, resource counters, and fallback status [2].

These measure different things. Iterating a provability predicate can add formal reflection depth without telling a controller which route is currently starved. Conversely, a controller can possess precise checked state facts without proving a long tower of nested statements about its own provability.

It is therefore useful to write a two-coordinate descriptor

$$\text{SA}(D) = (\text{lev}_{\text{ref}}(D), \text{lev}_{\text{state}}(D)),$$

where the first coordinate refers to the declared reflection schema and the second to checked architectural state description. This notation is a proposal for keeping the two ideas separate, not a claim that either coordinate measures consciousness.

18 Reflection can steer search without changing proof

Let the complete reflective state be

$$x = (\mathcal{B}, \mathcal{FB}, \text{H}, \text{Ref}, \text{Ctrl}, \dots).$$

The selector may use all coordinates. The verifier need not. If intrinsic acceptance is a function only of the frozen environment, admissible BANK dependencies, and the semantic certificate body, then changing reflection or controller state cannot change whether a fixed certificate is valid. This is the verification-neutrality principle of *What Checks the Proof?* [2].

Theorem 18.1 (Reflective extension preserves controlled-system form). *Suppose $(X, \{F_j\}, \pi)$ is a controlled discrete iterated system for an AMLD. Let R be a space of checked or checkable reflective-state records, and suppose each extended subsystem has a deterministic/effective update*

$$\hat{F}_j : X \times R \rightarrow X \times R.$$

Then the reflective machine is again a controlled discrete iterated system on $\hat{X} = X \times R$. If verifier acceptance remains independent of the reflective coordinate for fixed semantic certificate input, the reflective extension changes search policy without changing certificate semantics.

Proof. Closure under the maps $\hat{F}_j$ gives the controlled iterated-system representation immediately. Verifier neutrality follows from the stated interface independence: a coordinate that is not an argument of the acceptance predicate may affect which candidate is selected but not the acceptance value of a fixed candidate. $\square$

This theorem gives self-description an engineering role. Reflection is not asked to certify itself as truthful by fiat. It supplies checked features to a controller that still submits ordinary certificates to an unchanged verifier.

19 Main theorem III: checked reflective self-simulation and the minimal settlement quotient

The original checked-self-state theorem assumed the factorization

$$h^*(x) = g(s(x)).$$

That assumption is useful but too close to assuming the representation already contains exactly what the controller needs. The stronger result below replaces it by a local structural hypothesis: the checked self-state must correctly simulate the four generating updates of the same global machine. Ordinary induction then propagates that local agreement to every finite future role word.

Let

$$\Omega = \{\text{RUN}, P, R, I, C\}$$

and let

$$\tau : X \rightarrow \Omega$$

be the verifier-backed status label: $\tau(x) = \text{RUN}$ before terminalization, and $\tau(x) = k$ exactly on $\mathcal{C}_k$ after an accepted terminal certificate.

Let

$$s : X \rightarrow S$$

be a checker-backed self-description map available to the controller. A Level-2-style $s(x)$ may contain the current BANK fingerprint, active role, resource counters, recent verifier outcomes, branch dependencies, controller/fallback status, and any other checked variables needed to model the next global update. Assume the reflective machine contains abstract role maps

$$\bar{F}_j : S \rightarrow S, \quad j \in J,$$

and an abstract status map

$$\bar{\tau} : S \rightarrow \Omega.$$

Definition 19.1 (Checked generator self-simulation). The self-model $(S, s, \{\bar{F}_j\}, \bar{\tau})$ is *generator-exact* on a basin $B \subseteq X$ if, whenever $x \in B$ and the relevant successor remains in the basin,

$$\boxed{s(F_j x) = \bar{F}_j(s(x))} \quad \text{for every } j \in J,$$

and

$$\boxed{\tau(x) = \bar{\tau}(s(x))}.$$

The first equation says that one checked internal simulation step agrees with one actual role quantum after projection to self-state. The second says that the self-model does not invent or erase verifier terminal status.

For a word $w \in J^*$ define $\bar{F}_\epsilon = \text{id}$ and $\bar{F}_{wj} = \bar{F}_j \circ \bar{F}_w$.

Theorem 19.2 (Checked reflective self-simulation and settlement-quotient theorem). *Assume the complete deterministic four-role setup of Theorem 14.3, and suppose the checked self-model is generator-exact on a forward-closed basin B .*

(i) **Inductive self-simulation.** *For every $x \in B$ and every finite role word $w \in J^*$,*

$$\boxed{s(F_w x) = \bar{F}_w(s(x))}.$$

Consequently

$$\boxed{\tau(F_w x) = \bar{\tau}(\bar{F}_w(s(x)))}.$$

Thus local checked agreement on the four generators determines every finite abstract future of the complete P/R/I/C machine.

(ii) **Settlement-language factorization.** *Define the abstract outcome language*

$$\bar{\mathcal{L}}_k(q) := \{w \in J^* : \bar{\tau}(\bar{F}_w q) = k\}.$$

Then for every $x \in B$ and $k \in J$,

$$\boxed{\mathcal{L}_k(x) = \bar{\mathcal{L}}_k(s(x))}.$$

Hence all exact and truncated branchwise settlement depths factor through checked self-state:

$$r_k(x) = \bar{r}_k(s(x)), \quad d_{k,H}(x) = \bar{d}_{k,H}(s(x)).$$

In particular, wherever at least one terminal class is reachable, the scalar optimal settlement coordinate $h^* = -\min_k r_k$ factors through s as a derived conclusion rather than an independent hypothesis.

(iii) **Canonical settlement-future quotient.** *Define*

$$x \equiv_{\text{set}} y \iff \tau(F_w x) = \tau(F_w y) \text{ for every } w \in J^*.$$

Then $\equiv_{\text{set}}$ is an equivalence relation and a congruence for every generator F_j . Therefore the quotient

$$Q_{\text{set}} := B / \equiv_{\text{set}}$$

carries well-defined induced maps

$$\tilde{F}_j([x]) = [F_j x]$$

and an induced status map

$$\tilde{\tau}([x]) = \tau(x).$$

The quotient is an exact self-simulator of all future settlement labels.

(iv) **Minimality of exact Level-2 settlement state.** For every generator-exact self-model $s : B \rightarrow S$,

$$s(x) = s(y) \implies x \equiv_{\text{set}} y.$$

Thus every fiber of s lies inside one settlement-future equivalence class. If Q_{set} has finite cardinality N , then every exact self-model must realize at least N distinguishable self-states:

$$|s(B)| \geq N.$$

Any binary encoding of an exact self-model therefore requires at least

$$\lceil \log_2 N \rceil$$

bits in the worst case. Conversely, the quotient map $q : B \rightarrow Q_{\text{set}}$ itself is generator-exact, so this lower bound is attained at the level of abstract state count.

Proof. For (i), use induction on $|w|$. For the empty word,

$$s(F_\epsilon x) = s(x) = \bar{F}_\epsilon(s(x)).$$

Assume $s(F_w x) = \bar{F}_w(s(x))$. For any $j \in J$, generator exactness gives

$$s(F_{wj} x) = s(F_j(F_w x)) = \bar{F}_j(s(F_w x)) = \bar{F}_j(\bar{F}_w(s(x))) = \bar{F}_{wj}(s(x)).$$

This proves the first displayed identity for every finite word. Applying the status factorization $\tau = \bar{\tau} \circ s$ gives the second.

For (ii), a word w belongs to $\mathcal{L}_k(x)$ exactly when $\tau(F_w x) = k$. By (i), this is equivalent to

$$\bar{\tau}(\bar{F}_w(s(x))) = k,$$

which is exactly $w \in \bar{\mathcal{L}}_k(s(x))$. Equality of the languages implies equality of their shortest-word lengths and of every finite-horizon truncation. Taking the minimum over k gives the factorization of r^* and hence of h^* .

For (iii), reflexivity, symmetry, and transitivity follow from equality of the entire future status sequence indexed by words. If $x \equiv_{\text{set}} y$, then for any $j \in J$ and any continuation w ,

$$\tau(F_w(F_j x)) = \tau(F_{jw} x) = \tau(F_{jw} y) = \tau(F_w(F_j y)),$$

so $F_j x \equiv_{\text{set}} F_j y$. Hence each induced map $\tilde{F}_j$ is well-defined. Taking $w = \epsilon$ shows that equivalent states have the same current terminal label, so $\tilde{\tau}$ is well-defined as well.

For (iv), suppose $s(x) = s(y)$. By (i), for every word w ,

$$s(F_w x) = \bar{F}_w(s(x)) = \bar{F}_w(s(y)) = s(F_w y).$$

Applying $\bar{\tau}$ and generator exactness yields

$$\tau(F_w x) = \tau(F_w y),$$

so $x \equiv_{\text{set}} y$. Therefore two inequivalent quotient classes can never be merged into one self-state value, giving $|s(B)| \geq |Q_{\text{set}}| = N$. A binary code with fewer than $\lceil \log_2 N \rceil$ bits has fewer than N codewords, so it cannot distinguish all required self-states. Finally, congruence from (iii) makes the quotient projection satisfy

$$q(F_j x) = \tilde{F}_j(q(x)), \quad \tau(x) = \tilde{\tau}(q(x)),$$

so the quotient itself is exact. $\square$

Remark 19.3 (Relation to classical abstraction/minimization). Part (iv) has a deliberate Nerode-style flavor: states are identified exactly when no finite continuation of the role alphabet distinguishes their future output behavior [16]. State abstraction for control is also a classical subject [15]. The AMLD-specific content is the output being preserved—verifier-labeled P/R/I/C settlement futures—and the use of the quotient as the checked self-model of the same reflective settlement computer that chooses future role allocations. The general minimization idea is therefore cited rather than presented as new.

Corollary 19.4 (Finite reflective lookahead). *Assume Theorem 19.2, and suppose the finite-horizon restrictions of $\bar{F}_j$ and $\bar{\tau}$ are effectively computable by the controller. From the current checked state $s(x)$ the Level-2 ATP can enumerate every word $w \in J^{\leq H}$, compute*

$$\bar{\tau}(\bar{F}_w(s(x))),$$

and therefore compute the exact abstract quantities

$$\bar{d}_{k,H}(s(x)), \quad \bar{G}^H(s(x)), \quad \text{and, where defined, } \bar{\chi}_{ij \rightarrow k}^H(s(x)).$$

By Theorem 19.2, these equal the corresponding concrete finite-horizon settlement quantities. Thus a generator-exact Level-2 self-model permits the machine to evaluate counterfactual allocations of its own future computation without first executing all of those allocations in the external proof search.

Proof. There are finitely many words of length at most H . Compute their abstract trajectories recursively from $s(x)$ using the effective maps $\bar{F}_j$, read terminal labels using $\bar{\tau}$, and take the required minima and differences. Equality with the concrete quantities is Theorem 19.2(ii). $\square$

Corollary 19.5 (Checked self-state compass sufficiency: derived form). *Assume Theorem 19.2 on a basin where settlement is reachable, and define*

$$g(q) := -\min_{k \in J} \bar{r}_k(q).$$

Then

$$h^*(x) = g(s(x)).$$

If $\hat{g}$ satisfies, for every candidate successor y encountered,

$$|\hat{g}(s(y)) - g(s(y))| < \frac{1}{2},$$

then the controller

$$\pi_s(x) \in \arg \max_{j \in J} \hat{g}(s(F_j x))$$

selects a shortest-settlement action at every step and reaches settlement in $r^(x_0)$ steps.*

Proof. The factorization $h^* = g \circ s$ follows from Theorem 19.2(ii), rather than being assumed. Set $\hat{h}(y) = \hat{g}(s(y))$ and apply Theorem 12.1. $\square$

This derived corollary preserves the useful conclusion of the earlier Checked Self-State Compass Sufficiency theorem while strengthening its provenance: exact future sufficiency now follows from a checkable local self-simulation condition and induction over the four role generators.

20 Main theorem IV: settlement-growth algebra, N_H , and escape from the raw 4^H tree

The canonical quotient turns finite P/R/I/C lookahead into a finitely generated algebraic growth problem. Let

$$q : B \rightarrow Q_{\text{set}}$$

be the quotient map of Theorem 19.2, fix an initial state $x_0 \in B$, and let the free role monoid J^* act through the induced transformations $\tilde{F}_P, \tilde{F}_R, \tilde{F}_I, \tilde{F}_C$. The orbit of $q(x_0)$ is therefore a four-generator transformation-semigroup orbit. This section studies how many genuinely different settlement futures survive at depth H after the raw 4^H role words are identified by settlement-future equivalence.

Definition 20.1 (Settlement sphere growth and ball growth). For $H \in \mathbb{N}$, define the depth- H reachable quotient sphere

$$\mathcal{R}_H(x_0) := \{q(F_w x_0) : w \in J^H\}$$

and its cardinality

$$\boxed{N_H(x_0) := |\mathcal{R}_H(x_0)|.}$$

Also define the cumulative quotient ball

$$\mathcal{B}_H(x_0) := \bigcup_{h=0}^H \mathcal{R}_h(x_0), \quad B_H(x_0) := |\mathcal{B}_H(x_0)|.$$

Thus N_H is the sphere-growth function of the settlement quotient orbit and B_H is the corresponding ball-growth function.

Since there are exactly 4^H role words of length H ,

$$\boxed{1 \leq N_H(x_0) \leq 4^H,}$$

while

$$1 \leq B_H(x_0) \leq \sum_{h=0}^H 4^h = \frac{4^{H+1} - 1}{3}.$$

Neither N_H nor the quotient itself is assumed computable in an arbitrary infinite AMLD. On finite toy systems it can be exact; on realistic search systems it can be estimated or computed on a frozen finite basin. The point of the definition is to separate the raw role-word count from the number of settlement futures that actually remain distinguishable.

20.1 The four-coordinate profile and the product bound

For a continuation horizon $L \in \mathbb{N}$ and outcome $k \in J$, set

$$A_{k,L}(x) := \mathcal{L}_k(x) \cap J^{\leq L}$$

and define the truncated outcome-future profile

$$R_L(x) := (A_{P,L}(x), A_{R,L}(x), A_{I,L}(x), A_{C,L}(x)).$$

For the depth- H endpoints define

$$\mathcal{P}_{H,L}(x_0) := \{R_L(F_w x_0) : w \in J^H\}, \quad N_{H,L}(x_0) := |\mathcal{P}_{H,L}(x_0)|.$$

For each $k \in J$, let

$$m_{k;H,L} := \left| \{A_{k,L}(F_w x_0) : w \in J^H\} \right|$$

be the number of distinct values actually taken by the k th coordinate.

Proposition 20.2 (Realized-profile product bound). *For every H, L ,*

$$N_{H,L}(x_0) \leq m_{P;H,L} m_{R;H,L} m_{I;H,L} m_{C;H,L}.$$

Equality holds if and only if every combination of the four individually realized coordinate values is itself realized by at least one depth- H endpoint. Moreover,

$$N_{H,L}(x_0) \leq N_H(x_0) \leq 4^H,$$

$N_{H,L}$ is nondecreasing in L , and for each fixed H there exists a finite L_H such that

$$N_{H,L}(x_0) = N_H(x_0) \quad \text{for all } L \geq L_H.$$

Proof. The set $\mathcal{P}_{H,L}$ is a subset of the Cartesian product of its four coordinate projections. The cardinality of that Cartesian product is exactly the displayed product, proving the first inequality and the equality criterion. If two endpoint states are settlement-future equivalent, then their four truncated languages agree for every L , so a truncated profile cannot distinguish more classes than the full quotient; hence $N_{H,L} \leq N_H$. The inequality $N_H \leq 4^H$ follows because at most one quotient state is obtained from each of the 4^H words.

Increasing L can only add possible distinguishing continuations, so $N_{H,L}$ is nondecreasing. For fixed H there are only finitely many endpoint states. For every pair of inequivalent endpoint classes choose one finite role word that distinguishes their terminal labels; taking the maximum of the finitely many chosen distinguishing lengths gives L_H . At and beyond that horizon every pair of distinct full quotient classes is already separated by its truncated profile, so $N_{H,L} = N_H$. $\square$

Remark 20.3 (Why the product is not N_H). The expression

$$|A_{P,L}(x)| |A_{R,L}(x)| |A_{I,L}(x)| |A_{C,L}(x)|$$

counts ways of choosing one role word from each of four languages at one fixed state. It does not count future-equivalence classes. Even the product $m_{P;H,L} m_{R;H,L} m_{I;H,L} m_{C;H,L}$ is only an upper bound, because coordinate values can be correlated. For example, if the realized profiles are (A, B, C, D) , (A, B, E, D) , and (F, B, E, D) , then the four coordinate counts are 2, 1, 2, 1, whose product is 4, while only three complete profiles are realized.

Remark 20.4 (The crude product bound can be exponentially loose). The product bound is not itself an escape from exponential growth. Each coordinate count may grow rapidly with H , and the formal product may greatly exceed the already sharper universal bound $N_H \leq 4^H$. The computational question is therefore not whether a Cartesian-product upper bound exists, but how the *realized* quotient growth N_H compares asymptotically with 4^H .

20.2 Settlement Growth Escape Theorem

For $H \geq 1$, define the normalized settlement-growth exponent

$$a_H := \frac{\log N_H}{H \log 4}.$$

Because $1 \leq N_H \leq 4^H$, one always has $0 \leq a_H \leq 1$.

Theorem 20.5 (Settlement Growth Escape Theorem). *If*

$$\lambda := \limsup_{H \rightarrow \infty} \frac{\log N_H}{H \log 4} < 1,$$

then

$$\frac{N_H}{4^H} \rightarrow 0.$$

More quantitatively, for every c with $\lambda < c < 1$, there is H_0 such that

$$N_H \leq 4^{cH} \quad (H \geq H_0),$$

and hence

$$\frac{N_H}{4^H} \leq 4^{-(1-c)H}.$$

Thus the fraction of raw role words represented by distinct settlement-future classes decays exponentially.

Proof. Choose c with $\lambda < c < 1$. By the definition of limsup, $a_H \leq c$ for all sufficiently large H . Multiplying by $H \log 4$ and exponentiating gives $N_H \leq 4^{cH}$. Division by 4^H gives the displayed exponentially decaying upper bound. $\square$

Corollary 20.6 (Cumulative lookahead also escapes). *Under the hypothesis of Theorem 20.5,*

$$\frac{B_H}{\sum_{h=0}^H 4^h} \rightarrow 0.$$

Hence the number of distinct quotient states encountered through depth H is asymptotically negligible compared with the raw four-way prefix tree, provided quotient transitions are available without reconstructing the full raw tree first.

Proof. Choose c as in the theorem. Apart from finitely many initial layers,

$$B_H \leq C + \sum_{h=H_0}^H 4^{ch} = O(4^{cH}),$$

whereas $\sum_{h=0}^H 4^h = \Theta(4^H)$. Their ratio therefore tends to zero. $\square$

The last qualification is essential. A small quotient does not by itself prove that constructing the quotient is cheap. The theorem concerns the size of the compressed settlement geometry once the quotient or a certified representation of it is available. The engineering problem is to learn, discover, or maintain that quotient for less work than explicit raw expansion would require.

20.3 Subexponential and polynomial-scale regimes

Corollary 20.7 (Subexponential settlement growth). *If*

$$\log N_H = o(H),$$

equivalently $N_H = e^{o(H)}$, then $N_H/4^H \rightarrow 0$. This includes, but is not limited to, polynomial growth.

Proof. The hypothesis gives $a_H \rightarrow 0$, so Theorem 20.5 applies with $\lambda = 0$. □

For $H \geq 2$ define the polynomial-scale exponent

$$\beta_H := \frac{\log N_H}{\log H}.$$

Corollary 20.8 (Finite settlement-growth dimension). *If the finite limit*

$$\lim_{H \rightarrow \infty} \beta_H = d < \infty$$

exists, then

$$N_H = H^{d+o(1)}$$

and therefore

$$\frac{N_H}{4^H} \rightarrow 0.$$

In particular, the hypothesis does not assume the exact identity $N_H = H^d$; it asserts only that the logarithmic growth has a finite polynomial-scale limit.

Proof. The convergence $\beta_H \rightarrow d$ means

$$\log N_H = (d + o(1)) \log H.$$

Exponentiating gives $N_H = H^{d+o(1)}$. Since $\log H = o(H)$, it follows that $\log N_H = o(H)$, so Corollary 20.7 applies. □

This produces a useful hierarchy:

$$\begin{aligned} \limsup a_H < 1 &\implies \text{strictly sub-}4^H \text{ exponential rate,} \\ a_H \rightarrow 0 &\implies \text{subexponential settlement growth,} \\ \beta_H \rightarrow d < \infty &\implies N_H = H^{d+o(1)}, \\ N_H \text{ eventually bounded} &\implies \text{eventual finite saturation.} \end{aligned}$$

Every line implies $N_H/4^H \rightarrow 0$, but the lines make increasingly strong structural claims.

20.4 Nonstandard formulation: one hyperfinite algebra, with a uniform standard hypothesis

Let ${}^*\mathbb{N}$ be an elementary nonstandard extension containing unlimited hypernaturals, and transfer the standard sequence a_H to *a_K . The role words of hyperfinite length at most an unlimited K form an internal hyperfinite tree, while the transferred quotient orbit is an internal hyperfinite approximation to the settlement-growth algebra. As in Theorem 5.2, this is a semantic idealization, not an executable instruction to run an ordinary computer for K steps.

Theorem 20.9 (NSA settlement-growth escape criterion). *The following are equivalent:*

$$(i) \limsup_{H \rightarrow \infty} a_H < 1.$$

(ii) There exist standard constants $\delta > 0$ and $H_0 \in \mathbb{N}$ such that

$$a_H \leq 1 - \delta \quad \text{for every standard } H \geq H_0.$$

(iii) In the nonstandard extension, there exist the same standard $\delta > 0$ and H_0 such that

$${}^*a_K \leq 1 - \delta \quad \text{for every hypernatural } K \geq H_0.$$

In particular, the inequality then holds for every unlimited K .

Consequently any of these conditions implies $N_H/4^H \rightarrow 0$ by Theorem 20.5.

Proof. The equivalence of (i) and (ii) is the ordinary definition of a limsup lying strictly below 1: choose a standard gap δ smaller than $1 - \limsup a_H$. The equivalence of (ii) and (iii) is Transfer applied to the eventually uniform inequality. The final implication is Theorem 20.5. $\square$

Corollary 20.10 (NSA characterization when an ordinary growth limit exists). *If $a_H \rightarrow a$, then for every unlimited K ,*

$${}^*a_K \approx a, \quad \text{st}({}^*a_K) = a.$$

If instead $\beta_H \rightarrow d < \infty$, then for every unlimited K ,

$${}^*\beta_K \approx d, \quad \text{st}({}^*\beta_K) = d,$$

and therefore $N_H = H^{d+o(1)}$ and $N_H/4^H \rightarrow 0$.

Proof. These are the standard nonstandard characterizations of convergence applied to the transferred sequences. The polynomial-scale conclusion is Corollary 20.8. $\square$

Remark 20.11 (Why “one unlimited K ” is not enough). A favorable value of *a_K at one selected unlimited hypernatural does not by itself control the standard limsup. The robust NSA statement requires a standard positive gap from 1 that holds eventually and therefore, by Transfer, at every sufficiently large hypernatural. This prevents a sparse favorable subsequence from being mistaken for a global asymptotic escape result.

20.5 Catalysis as a source of quotient distinctions

The settlement-semigroup theorem supplies a direct algebraic interpretation of these growth counts. Its residual law

$$\mathcal{L}_k(F_u x) = u^{-1} \mathcal{L}_k(x)$$

says that the state reached after history u carries the left residual of each outcome language. Thus a settlement-future class can be represented by the four residual outcome languages, and truncated $R_L(x)$ is their finite-horizon shadow.

Catalysis does not force N_H to move monotonically in one direction. If a cross-role prerequisite records settlement-relevant history—for example, an R action must first create a BANK condition that later allows P to settle—then previously equivalent residual futures may split, raising N_H . Conversely, if catalysis causes many histories to enter the same absorbing settlement class rapidly, terminal absorption can collapse many futures and lower N_H . The net N_H therefore measures both *distinction creation* from order/history dependence and *distinction destruction* from merging and absorption.

For a fixed horizon one may define an off-diagonal catalysis signature

$$\kappa_H(x) := (\mathbf{1}\{j \text{ is } k\text{-catalytic at } x\})_{j \neq k}.$$

Whenever the horizon is long enough to witness the relevant shortest routes, distinct verified catalysis signatures provide a lower bound on the number of future profiles that a controller must distinguish. Richer signatures that include ablation indicators and order defects can only refine this diagnostic. This is a measurement consequence of the settlement-semigroup theorem, not a claim that catalysis alone determines the growth exponent of N_H .

Remark 20.12 (Prior-art boundary). Growth functions of finitely generated groups, monoids, semigroups, and automata are classical subjects, as are residual-language and Nerode-style state minimization. The AMLD-specific research object here is the growth of a verifier-labeled P/R/I/C settlement quotient, its relation to cross-role catalysis and terminal absorption, and its use as an online criterion for whether checked reflective lookahead is compressing the raw role tree enough to justify its cost. Historical priority for the general algebraic ingredients is not claimed.

21 Three conjectures suggested by the strengthened theory

The theorems above establish identities and sufficiency conditions. They do not establish that the corresponding phenomena are common in difficult mathematics. The following claims are therefore explicitly conjectural and should be tested on withheld theorem/refutation/independence/inconsistency instances before being invoked on open mathematics.

Conjecture 21.1 (Cross-role catalysis on hard settlement trajectories). *There exist leakage-resistant hard-instance distributions and horizons H for which off-diagonal shortest-route progress occurs with nonnegligible frequency:*

$$\Pr[\exists j \neq k : G_{jk}^H(x) = 1] > 0,$$

and the frequency or magnitude of such effects is strictly reduced by a matched ablation that prevents verified cross-role BANK reuse. Equivalently, on some difficult instances the shortest settlement of one outcome class genuinely depends on computation performed by a different nominal role.

Conjecture 21.2 (Noncommutative scheduling advantage). *There exist target families on which the order defects $\chi_{ij \rightarrow k}^H(x)$ are nonzero on a positive fraction of relevant states and a controller allowed to exploit their signs and magnitudes achieves strictly smaller expected verifier calls to accepted settlement than a matched controller that is invariant under swapping the corresponding role orders.*

Conjecture 21.3 (Reflective settlement compression and transfer). *For useful theorem-settlement families there exist checker-backed self-state maps s that are dramatically smaller than the raw implementation state while preserving, exactly or approximately, enough of the finite-horizon settlement quotient to recover G^H or its shortest-route entries on held-out searches. A controller using such checked reflective state then improves end-to-end verified settlement under fixed resources relative to otherwise matched nonreflective or state-ablated baselines.*

These conjectures state what would make the architecture useful for difficult conjectures. Conjecture 21.1 asks whether the federation is more than parallel redundancy. Conjecture 21.2 asks whether its IFS/semigroup order structure contains exploitable information. Conjecture 21.3 asks whether Level-2 self-description can compress the relevant future geometry enough to make self-simulation computationally worthwhile.

22 Fairness, failure, and the independence route

Adaptive control creates a danger: a controller may learn that some histories correlate with independence and then silently treat prolonged proof failure as a metatheorem. That is not allowed.

Automatic Metalogical Deciders, Part Two makes the canonical rule intentionally conservative: P, R, and I receive equal opportunities under the standard scheduler, and long failure histories are not converted into logical status [8]. Adaptive allocation remains legitimate as an experimental policy provided two boundaries remain clear:

1. only an explicit accepted independence certificate licenses the independence outcome; and
2. any liveness guarantee inherited from complete baseline procedures is protected by a fairness floor or persistent fallback.

This distinction is compatible with learning from failure. A state descriptor may record “P has spent N verifier calls without settlement.” Historical data may show that such a state predicts a higher conditional chance that an I-action will settle soon. The controller may use that information to rank actions. But the statement

“P and R have failed for a long time”

is still not itself the certificate

$$\Gamma \not\vdash c \quad \text{and} \quad \Gamma \not\vdash \neg c.$$

23 Why negative experiments belong in the theory

A compass theory that only records positive navigation stories is not a scientific control theory. Recent work on compass/density hypotheses supplies useful negative examples. In one frozen DATA 4.1-Q benchmark, a preregistered shell-concentration hypothesis failed, and a strong practical physics-density advantage was denied because the measured mean-rank improvement was only 0.552%, below the preregistered 10% threshold [9]. These outcomes do not show that every physics-inspired or density-based feature is useless. They show that architectural plausibility is not evidence of practical value.

The lesson for the present program is methodological. Theorems should establish safety, invariants, and idealized implications. Experiments should determine whether proposed representations and learned coordinates transfer to new search states. Negative results should eliminate mechanisms, revise hypotheses, or identify missing state variables rather than being hidden behind new terminology.

24 Returning to the business of settling conjectures

The strengthened theory is worthwhile only if it changes how difficult targets are attacked. Its purpose is not to create more terminology around proof search. It creates measurable objects— $\mathcal{L}_k$, r_k , $\Delta_{j \rightarrow k}$, G^H , χ^H , and Q_{set} —that can be tested against end-to-end verified settlement.

24.1 A three-stage settlement program

Stage A: solved but hidden targets. Use theorems whose certificates are known to the experimenter but withheld from the search machine. Freeze Γ , target, verifier, role-internal policies, candidate rules,

resource budget, BANK policy, and random seeds before evaluation. On manageable state graphs compute exact or near-exact r_k and $d_{k,H}$ retrospectively. From these labels compute the exact cross-agent progress matrices G^H , order defects χ^H , role-ablation penalties $\Delta_{j \rightarrow k}$, and empirical settlement-future equivalence classes. Stage A is where the new theorems can be falsified as implementations: if the measured software transitions do not satisfy the formal identities, the implementation is not the mathematical object claimed here.

Stage B: mixed sealed settlement instances. Construct a sealed mixture containing theorem, refutation, independence, and inconsistent-base cases with certificate classes declared in advance. Compare unbiased round-robin, standard professional ATP baselines where appropriate, BANKed search, scalar compass search, cross-agent-matrix guidance, order-aware guidance, and reflective self-simulation guidance. Include matched ablations that block cross-role BANK reuse, erase role order, or collapse checked self-state features. The primary endpoint is verifier calls or wall-clock time to the first accepted terminal certificate. Timeouts remain UNKNOWN.

Stage C: genuinely difficult mathematics. Only after the controller survives leakage-resistant Stages A and B should it be aimed at problems whose status is not known to the system builder. The formal environment must still be frozen. For an open conjecture, a long run that produces no certificate is simply an unsuccessful run. A positive claim requires an independently checkable proof, refutation, independence/metacertificate, or inconsistency certificate of the declared kind. The control theory may change where the machine looks; it never changes what the verifier accepts.

24.2 What to measure

The primary measurement remains end-to-end verified settlement under fixed resources. Diagnostics should now include:

- scalar error or ranking loss for $\hat{h}$ versus retrospective h^* where computable;
- branchwise error for $\hat{d}_{k,H}$ and the fraction of states satisfying the half-step and quarter-step sufficient conditions;
- exact recovery rate for the matrix G^H and for signs/values of the order defects $\chi_{ij \rightarrow k}^H$;
- off-diagonal catalytic frequency $G_{jk}^H = 1$ for $j \neq k$;
- role-ablation penalties $\Delta_{j \rightarrow k}$, especially off-diagonal values and $+\infty$ indispensability events;
- P/R/I/C allocation history, role-order history, and fairness-floor usage;
- number and downstream marginal value of BANK deposits, including whether a deposit changes another role's shortest settlement distance;
- empirical semiconjugacy violations $s(F_j x) \neq \bar{F}_j(s(x))$ and terminal-label mismatches;
- collisions $s(x) = s(y)$ for which short continuation words distinguish future settlement labels, as finite witnesses that the proposed self-state is too coarse;
- the number of empirical settlement-future quotient classes at increasing horizons and the compression ratio between raw state and checked self-state;
- verifier calls, wall-clock time, peak memory, and restart/checkpoint overhead;
- ablations removing self-state features, FUTUREBANK features, BANK reuse, order information, or compass guidance;
- negative controls that preserve candidate coverage while randomizing role ordering or ranking.

24.3 The theorem-driven experimental question

Let π_0 be a frozen baseline policy, π_G a controller using estimated cross-agent geometry $(\hat{G}, \hat{\chi})$, and π_S a reflective policy using a checker-backed self-model. On a preregistered target distribution and resource budget B , compare

$$\Pr(\tau_{\text{settle}} \leq B \mid \pi_0), \quad \Pr(\tau_{\text{settle}} \leq B \mid \pi_G), \quad \Pr(\tau_{\text{settle}} \leq B \mid \pi_S),$$

where settlement means an accepted terminal certificate. Secondary analyses ask whether any gain is mediated by off-diagonal catalysis, order-sensitive scheduling, earlier high-value BANK deposits, or compression of raw search state into a self-model that preserves the relevant settlement futures.

The desired end state is not a machine with a sophisticated vocabulary for itself. It is a verifier-gated machine that can model the consequences of allocating its own future computation, exploit cross-role mathematical work when it matters, and thereby produce more independently checkable settlements under the same resource budget.

25 A compact theorem map

Structural and classical foundations.

Result	Status	Meaning
BANK–SIC representation	Theorem	AMLD BANK evolution becomes a SIC trajectory after forgetting record metadata.
Γ seed correspondence	Corollary	Γ is the admitted initial mathematical state in both SIC and AMLD views.
Proof-Horizon–NSA equivalence	Theorem	Ordinary horizon halting, standard derivability, and proof existence below an unlimited nonstandard cost bound agree on standard targets.
Conditional settlement-ATP	Theorem	Fair complete coverage of declared certificate families makes global settlement a target-search ATP for a settlement proposition.
Controlled IFS representation	Theorem	The complete AMLD evolves by selected global state maps F_j .
Full-state noncontractivity / quotient contractivization	Theorems	Distinct frozen terminals obstruct full-state contraction; terminal quotienting can yield a contraction on a finite-settling basin.
Controlled Abel–Bellman	Classical foundation	$h^* = -r^*$ obeys the Bellman/Abel one-step optimality identity.
Half-step compass	Classical foundation	Uniform successor error $< 1/2$ recovers shortest-settlement actions in the unit-cost model.

Strengthened main sequence.

Result	Status	Meaning
Settlement-semigroup and catalysis	Main theorem	Outcome-labeled future languages obey a residual law; shortest-word depths yield exact cross-role catalysis, ablation penalties, and order-sensitive settlement defects.
Quarter-step cross-agent geometry	Main theorem	Two-point prediction errors totaling $< 1/2$ recover the exact integer 4×4 progress matrix; symmetric $< 1/4$ errors suffice, and pairwise role-order defects are exactly recoverable.
Reflective self-simulation and settlement quotient	Main theorem	Generator-exact checked self-state simulates every finite P/R/I/C future by induction, factors settlement geometry, and induces a canonical minimal future-preserving quotient.
Settlement-growth product bound	Proposition	$N_{H,L}$ counts realized four-coordinate outcome-future profiles and is bounded by the product of the four coordinate-value counts, with equality only when all coordinate combinations occur.
Settlement Growth Escape	Main theorem	If $\limsup \log N_H / (H \log 4) < 1$, then $N_H / 4^H \rightarrow 0$ exponentially; the cumulative quotient ball is likewise negligible relative to the raw prefix tree.
NSA settlement-growth criterion	Main theorem	A standard eventual gap below exponent 1 transfers to every sufficiently large hypernatural horizon; ordinary growth limits can be read by standard part at unlimited horizons.
Finite reflective lookahead	Corollary	An effective generator-exact self-model can compute finite counterfactual role-allocation futures internally.
Checked self-state compass sufficiency	Derived corollary	The old factorization hypothesis follows from generator-exact self-simulation; half-step accuracy then gives optimal greedy settlement control.
Cross-role catalysis	Conjecture	Hard settlements frequently contain shortest-route moves by a role different from the eventual terminal label, with BANK reuse contributing to the effect.
Noncommutative scheduling advantage	Conjecture	Role order defects occur often enough that order-aware control improves verified settlement under fixed resources.
Reflective settlement compression	Conjecture	A compact checked self-state preserves enough settlement-future geometry to improve held-out verified settlement.

26 Relation to the surrounding research program

The present manuscript is related to, but not dependent on, several larger texts and working directions.

What Checks the Proof? supplies the verification-centered architecture: Γ , BANK, FUTUREBANK, P/R/I/C, intrinsic verification, fair fallback, and the distinction between search and certification [2]. Its job is foundational.

Depths of Induction and Formalized Self-Awareness supplies SICs and ATPs as staged formal computers, together with tagging, self-models, self-theories, reflection sequences, and formal self-awareness levels [3]. Its role here is to make the search machine itself an object of formal mathematics.

The *Universal Proof-Horizon Operator* supplies the exhaustive positive-side baseline: under an effectively

proper proof cost, a partial computable operator finds a canonical minimum-cost proof exactly on theorem inputs. The NSA chapter of *What Checks the Proof?* supplies a distinct semantic idealization in which an unlimited bound covers every standard finite proof code. Theorem 5.2 connects these formulations while preserving their computational boundary: the nonstandard bound dominates all standard horizons but is not a standard executable cutoff [1, 2].

DATA 3.0 supplies the dynamical backbone: complete state, typed verified BANK, state transformations, controlled/place-dependent iteration, transformation semigroups, target fibers, terminal fixed points, settlement horizons, and the retrospective Abel coordinate [4].

Proof Compass Theory: Learnability, Shrinkage, and Control supplies explicit sufficient conditions under which learned navigation can outperform uninformed ordering while retaining safe fallback; it also records why unrestricted compass learnability is false without visible signal [5]. AMLD compass-control work supplies the optimal-control interpretation: value functions, distance-to-settlement, Lyapunov/Abel coordinates, Bellman feedback, and the operational value of theorem-shell changes [6].

Automatic Metalogical Deciders, Part Two supplies scheduling discipline and negative results about naive resource heuristics, especially the rule that failure is information for control but never a substitute for a certificate [8]. The *Reflective Compass Control Research Program* supplies a nearby implementation research agenda: certified self-state, meta-policy selection, conservative fallback, and native verifier sovereignty [7].

External prior work helps delimit what is and is not being claimed here. Bellman supplies the classical optimal-control recurrence [13]; Farahmand supplies action-gap robustness [14]; Li, Walsh, and Littman supply a unified state-abstraction framework for MDPs [15]; Nerode supplies the canonical future-equivalence/minimal-state ancestry [16]; Sonoda, Akiyama, and Uezato explicitly model agentic theorem proving with finite-horizon MDP/Bellman structure [17]; and Ahuja and Gour study multi-agent theorem proving as coordinated allocation over shared proof infrastructure [18]. The strengthened sequence therefore focuses on the AMLD-specific synthesis rather than claiming those general ideas as new.

The working title *What Watches the Search?* can naturally develop the reflection/control side further. A specialized working title, *Initial Segments of Tegmark’s Self-Aware Substructures: Formal Self-Aware Automated Theorem Provers*, can isolate the philosophical and order-theoretic questions without treating formal self-reference as consciousness. A later *Techniques to Settle Conjectures* can then reverse the emphasis: less architecture, more theorem targets, ablation-tested search strategies, and archived verifier certificates.

The direction of travel should be deliberate:

$$\boxed{\text{build the correctness boundary}} \rightarrow \boxed{\text{understand and control the search}} \rightarrow \boxed{\text{settle mathematics.}}$$

27 Conclusion

The manuscript now separates the classical foundations from the AMLD-specific research object. Bellman optimality still gives the exact shortest-settlement recurrence, and the half-step theorem still gives a sufficient robustness margin for greedy action selection. Those results remain important, but they are no longer asked to carry the originality claim.

The stronger theory begins when the complete verifier-gated P/R/I/C federation is treated as one evolving discrete computer. Its four role updates generate a semigroup of global state transformations

and, for each verifier terminal class, an outcome-labeled settlement language. The residual law

$$\mathcal{L}_k(F_u x) = u^{-1} \mathcal{L}_k(x)$$

turns a current role history into a precise description of the remaining future schedules that can settle each outcome. The branchwise shortest-word depths then define cross-role catalysis, role-ablation penalties, and order defects. These quantities ask questions a scalar value cannot ask: whether R is necessary on every shortest P-route, whether P-before-I is better than I-before-P for an eventual independence certificate, or whether removing one nominally “wrong” role destroys the shortest settlement geometry.

The quarter-step theorem converts this structure into a measurable compass target. A Level-2 ATP need not recover an unknowable global proof landscape perfectly. On a finite horizon it can estimate four branchwise settlement depths and, if the two-point errors are small enough, recover the exact integer 4×4 progress matrix and pairwise order defects. The useful object is therefore not only “distance to settlement” but a local discrete interaction geometry of the entire P/R/I/C system.

The reflective theorem supplies the strongest conceptual change. Instead of assuming that checked self-state is sufficient for optimal control, the machine is required to model the four generators of its own evolution. If

$$s \circ F_j = \bar{F}_j \circ s$$

for the four roles and terminal status also factors through s , ordinary induction on role words gives

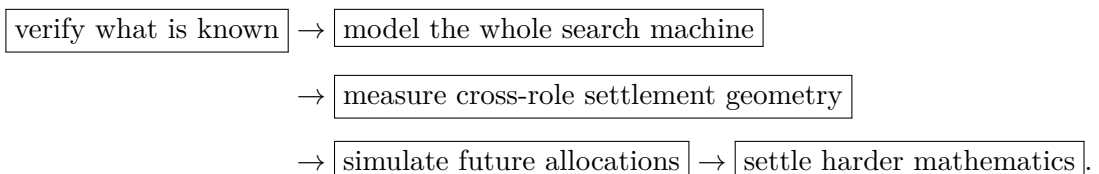
$$s \circ F_w = \bar{F}_w \circ s$$

for every finite future schedule w . The Level-2 ATP can then use a checked model of itself to evaluate counterfactual allocations of its own future computation. The settlement-future quotient further identifies the minimum exact state distinction required to preserve every verifier-labeled P/R/I/C future and yields a corresponding lower bound on exact self-state information.

None of this settles a hard mathematical conjecture by architecture alone. The proved results identify quantities, invariants, sufficiency conditions, and exact consequences. The three new conjectures ask whether the phenomena that would make the framework practically powerful—cross-role catalysis, exploitable noncommutativity, and reflective compression—actually occur on difficult held-out mathematics. That question belongs to sealed experiments. If the answer is positive, the ultimate use is concrete: make the whole settlement computer better at deciding which internal mathematical mode should act next, while leaving the external verifier sovereign.

The settlement-growth extension sharpens what “compression” must mean. It is not enough for a quotient to have fewer states on one example. The decisive asymptotic comparison is the realized growth N_H against the raw 4^H role tree. A limsup exponent strictly below 1 already implies an exponentially vanishing quotient fraction; subexponential or polynomial-scale growth is stronger still. The NSA formulation packages the same eventual inequality into a hyperfinite algebra only after a uniform standard gap has been established, preserving the distinction between semantic idealization and executable search.

The direction of travel is therefore sharper than before:



Every positive mathematical claim still ends at an independently checkable certificate.

A Exploratory synthetic N_H experiment catalogue

This appendix records the exploratory toy-system results used to motivate Section 20. They are *synthetic diagnostics*, not evidence about the frequency of compression or catalysis in a production AMLD or in difficult mathematics. The figures below are the session-level pilot outputs obtained while developing the definitions; publication-grade use should freeze the generator code, random seeds, exact state semantics, and analysis script and then replicate the table from those artifacts.

The common operational comparison at horizon $H = 6$ is the raw role-word count

$$4^6 = 4096.$$

The compression ratio is $N_6/4096$ and the raw-to-quotient factor is $4096/N_6$.

A.1 Experiment 1: geometry stress test

The first pilot used 60 deterministic four-role toy AMLDs, divided into 20 nearly tree-like, 20 moderately merging, and 20 strongly compressible systems. The exploratory medians at $H = 6$ were approximately

$$3770.5, \quad 943, \quad 23,$$

corresponding to quotient fractions 0.921, 0.230, and 0.0056. The associated compression factors were about $1.09\times$, $4.34\times$, and $178\times$.

A later cleaner controlled generator produced medians

$$3324, \quad 1036, \quad 31,$$

with quotient fractions approximately 0.812, 0.253, and 0.00757, or compression factors about $1.23\times$, $3.95\times$, and $132\times$. The numerical values moved with the toy generator, but the qualitative ordering—near-tree $\gg$ moderate $\gg$ strongly compressible—survived.

A.2 Experiment 2: pure settlement-label permutation

On matched systems, permuting the names P, R, I, C while leaving the transition structure unchanged produced exactly zero change in N_H . This is a required sanity check: the quotient counts settlement behavior, not the typography of the terminal labels.

A.3 Experiment 3: one possible outcome with symmetric agents

When only one settlement class was enabled at a time while the four roles remained statistically symmetric, the reported $H = 6$ medians were

sole outcome	P	R	I	C
N_6	1038.5	1039.5	1035.5	1038

The spread was below one half percent. Outcome identity by itself therefore had little effect in this matched toy design.

A.4 Experiment 4: unequal settlement accessibility

When the four outcomes were assigned different settlement opportunity rates, approximately

$$P : 0.20, \quad R : 0.10, \quad I : 0.05, \quad C : 0.02,$$

the reported results were

outcome	terminal fraction at $H = 6$	N_6	$4096/N_6$
P	20.6%	2233	1.83
R	11.5%	2375	1.72
I	5.6%	2463	1.66
C	1.6%	2542	1.61

Greater settlement accessibility produced more terminal absorption and, in this toy family, smaller N_6 .

A.5 Experiment 5: settlement-accessibility sweep

A broader accessibility sweep gave

q	.005	.01	.02	.05	.10	.20	.40
median(N_6)	2566.5	2564	2543	2471	2368.5	2225.5	1799

which was monotone over the tested grid. The result is descriptive for this generator, not a theorem that N_H must decrease with settlement probability in every AMLD.

A.6 Experiment 6: outcome diversity at approximately fixed total settlement rate

Matched systems with one, two, or four possible terminal labels, including strongly P -dominant and strongly C -dominant variants, produced medians in the narrow range

$$1186.5 \leq N_6 \leq 1192.5.$$

Thus outcome-label diversity alone was weak relative to branch merging and terminal accessibility in this pilot.

A.7 Experiment 7: cross-role catalysis

A matched toy family compared direct settlement with a cross-role prerequisite, such as an R -generated condition that had to occur before a later P settlement. The reported medians were approximately

$$N_6 \approx 474 \quad (\text{direct}), \quad N_6 \approx 797 \quad (\text{cross-role prerequisite}).$$

The corresponding compression factors were about $8.65\times$ and $5.14\times$. In this construction the prerequisite increased effective future complexity by roughly

$$\frac{797 - 474}{474} \approx 0.68.$$

This does not contradict the possibility that catalysis can lower N_H in other systems: cross-role memory can split residual futures, while rapid successful settlement can merge them through absorption. The settlement-semigroup theorem predicts the relevant residual-language distinctions but does not impose a universal sign on the net change in N_H .

A.8 Recommended confirmatory measurements

For every future frozen toy or real AMLD experiment, record at each horizon

$$N_H, \quad \frac{N_H}{4^H}, \quad \frac{N_H}{N_{H-1}}, \quad \frac{\log N_H}{H \log 4},$$

along with settlement rate, outcome distribution, cross-role catalysis frequency, ablation frequency, order-defect frequency, and quotient-construction cost. The first four describe growth; the remaining variables help explain *why* that growth is large or small. The decisive practical comparison must include the cost of discovering or maintaining the quotient, not only the smaller number of quotient states after construction.

AI Integrity Statement

AI assistance was used to synthesize earlier manuscripts, test proof ideas, draft exposition, perform the Version 1.8 prior-art-oriented restructuring and the Version 2.2 settlement-growth extension, and produce the typeset manuscript. Mathematical claims labeled as theorems were checked against their stated hypotheses during preparation, but they remain appropriate subjects for independent scholarly review and, where feasible, formal verification. Empirical claims have not been promoted to theorems. No open conjecture is reported as settled without an independently checkable certificate.

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